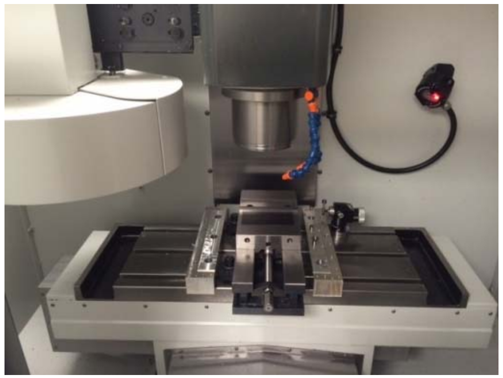
# Advanced Manufacturing Technology & Prototyping (MDP494)

## Lecture 1

## Revision: Milling process



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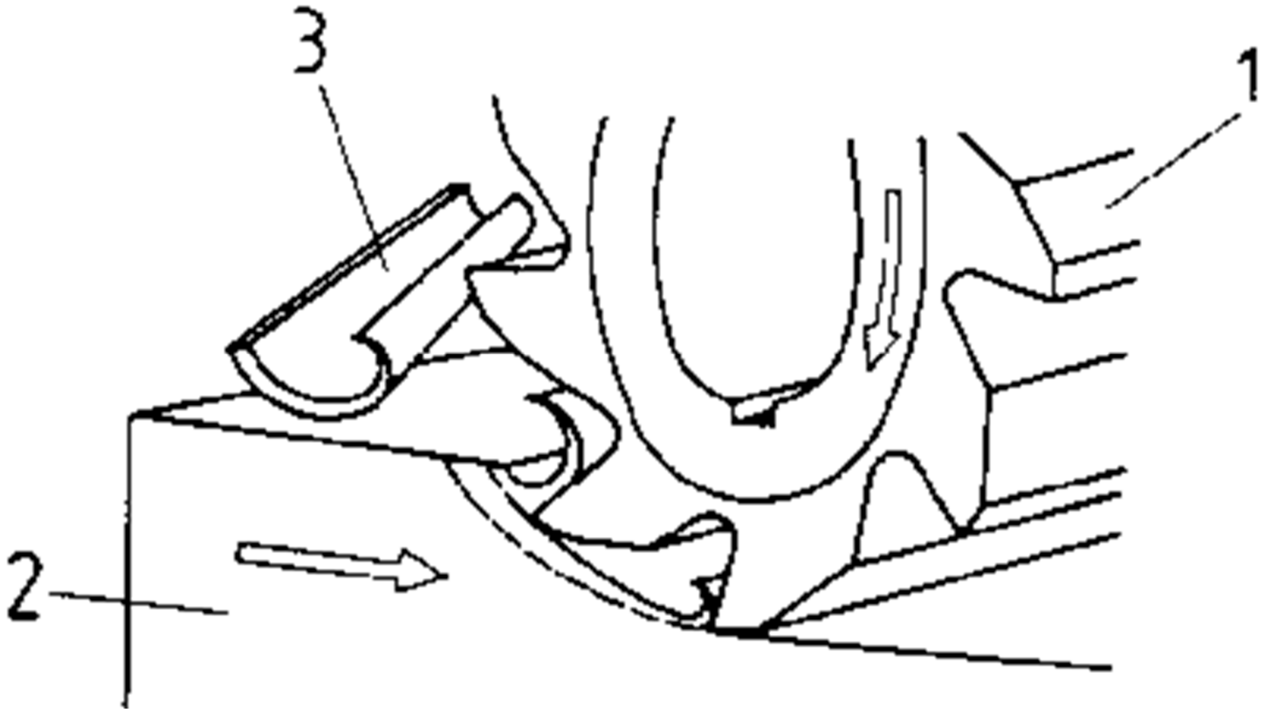
# MILLING PROCESS



# Objective

- **Study the milling process to understand its basic principle, classification, procedure, application and equipment; tools and machine.**

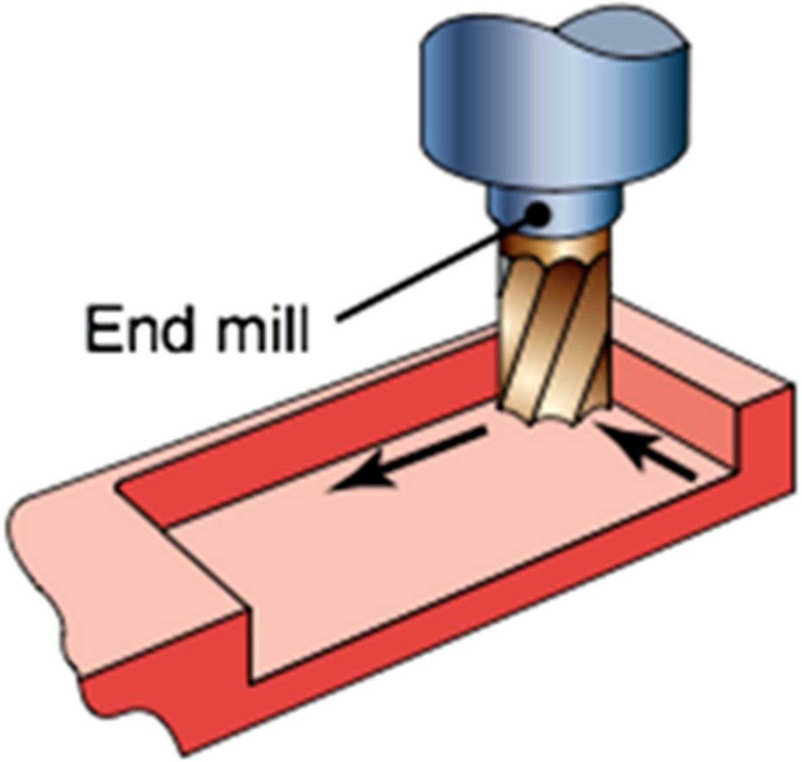
# Objective

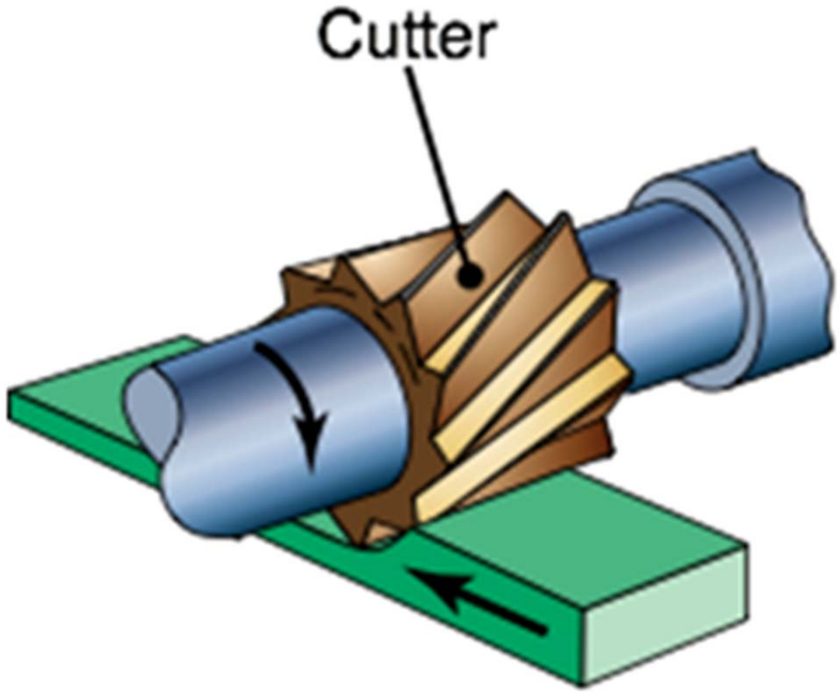
- **Identify the cutting parameters (speed, feed and depth of cut) for the milling process.**
- **Identify the basic parts of the horizontal and vertical milling machines.**
- **Know the different ways of tool and work piece fixation.**
- **Identify the different operation carried out on the milling machine.**
- **Machining time calculation.**

# Definition

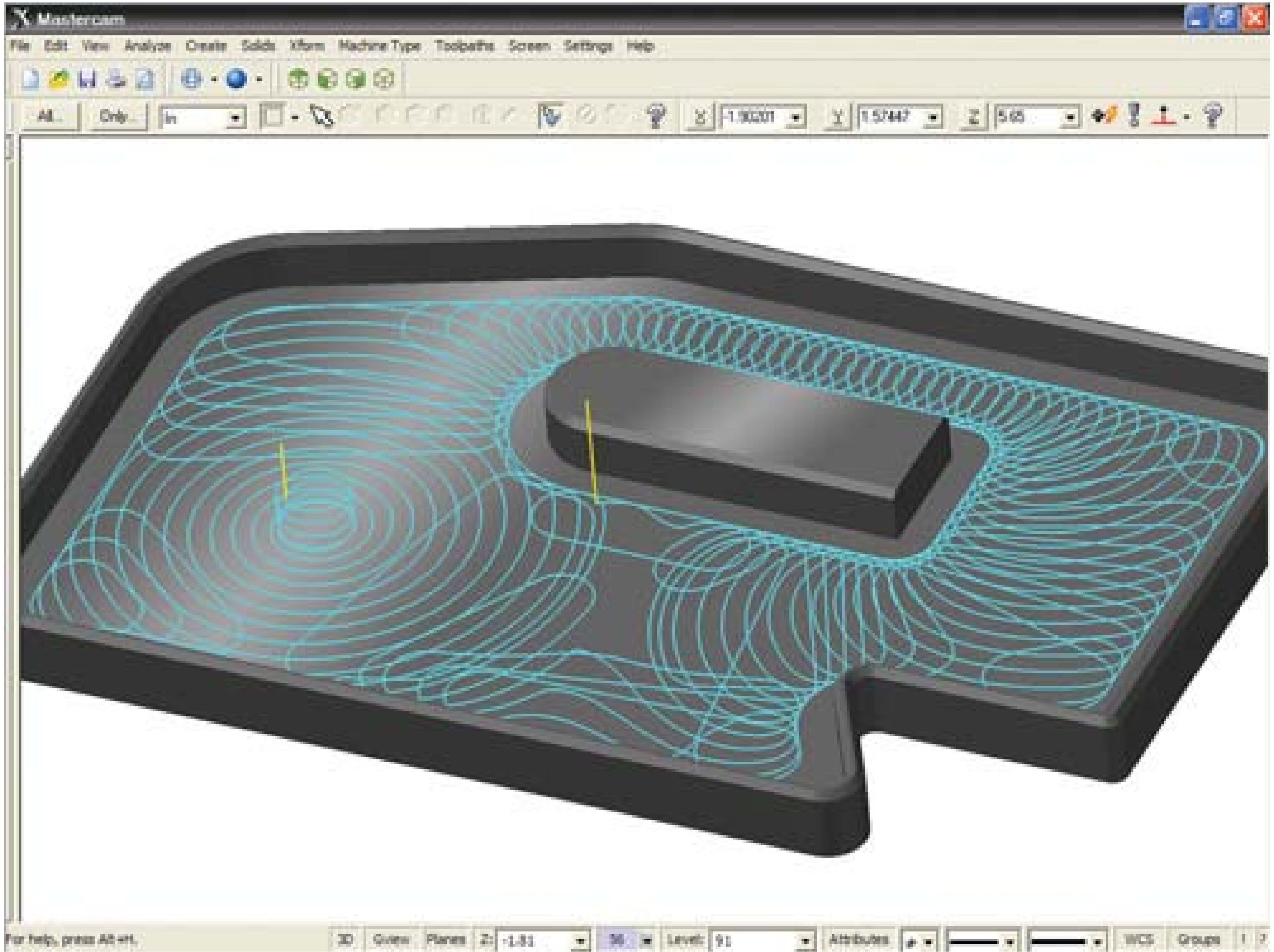
- **Milling is a metal cutting process used for the generation of flat, curved, or irregular surfaces.**
- **Milling operations are performed on a horizontal or vertical milling Machine.**
- **Achieved by *rotating* the *tool* and holding a *stationary* *work piece* against it with a certain depth of cut.**
- **The milling cutter containing a number of cutting edges**

# Process

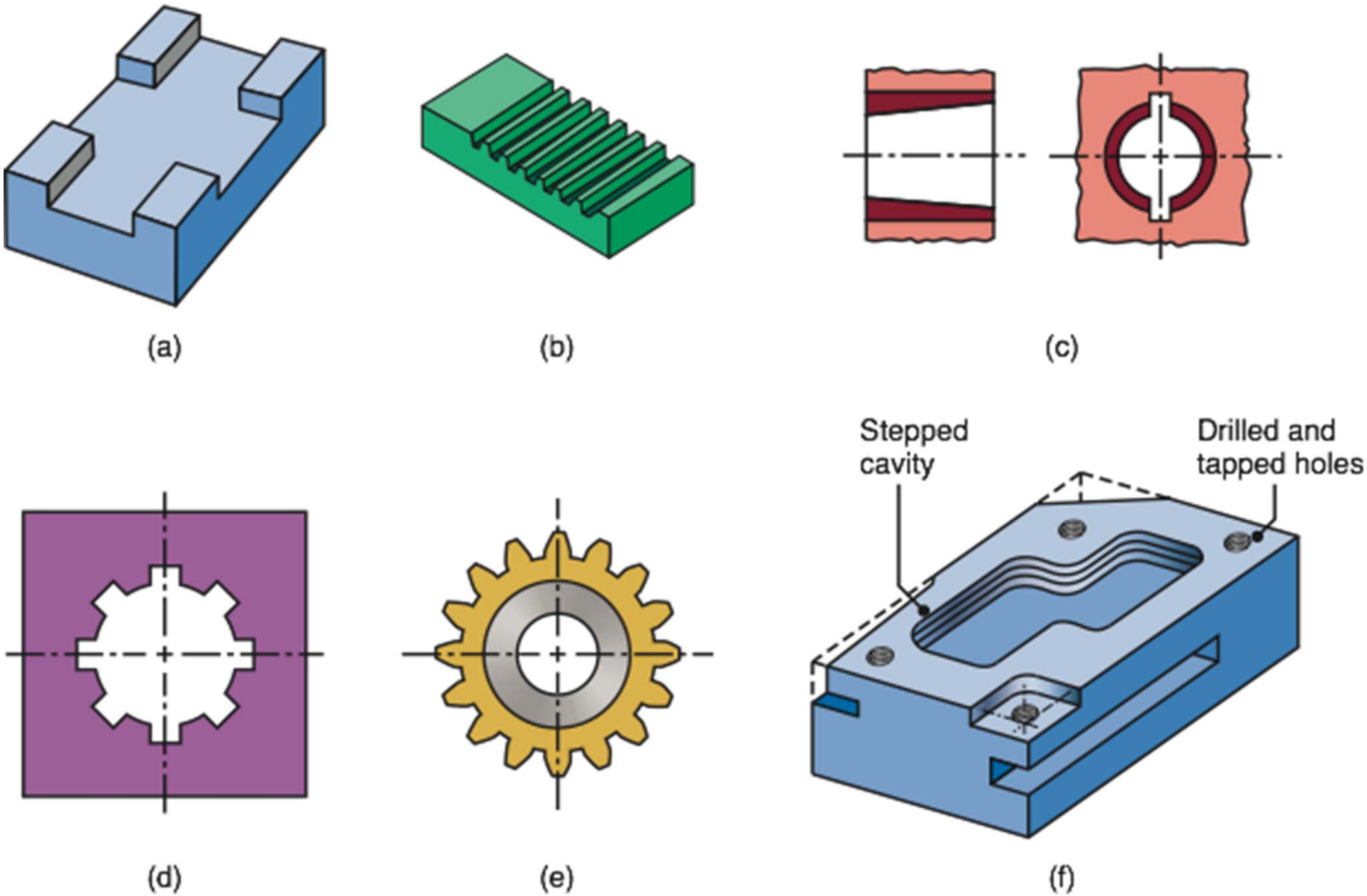


# Typical parts and shapes produced by milling



# Typical parts and shapes produced by milling



# Cutting conditions

## 1) Cutting Speed (N): سرعة القطع

**Rotational Speed in rpm**

**Is done by the cutting tool**

## 2) Feed (f): التغذية

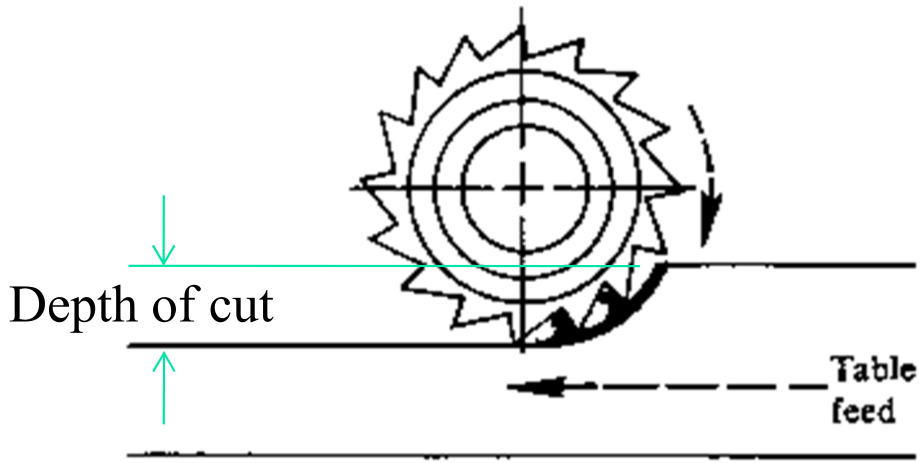
**This is represented in either mm/min or mm/rev**

**Is done by the table**

## 3) Depth of cut (a): عمق القطع

**This presented in mm**

**Is done by the table**



# Milling Machines

# Milling machines

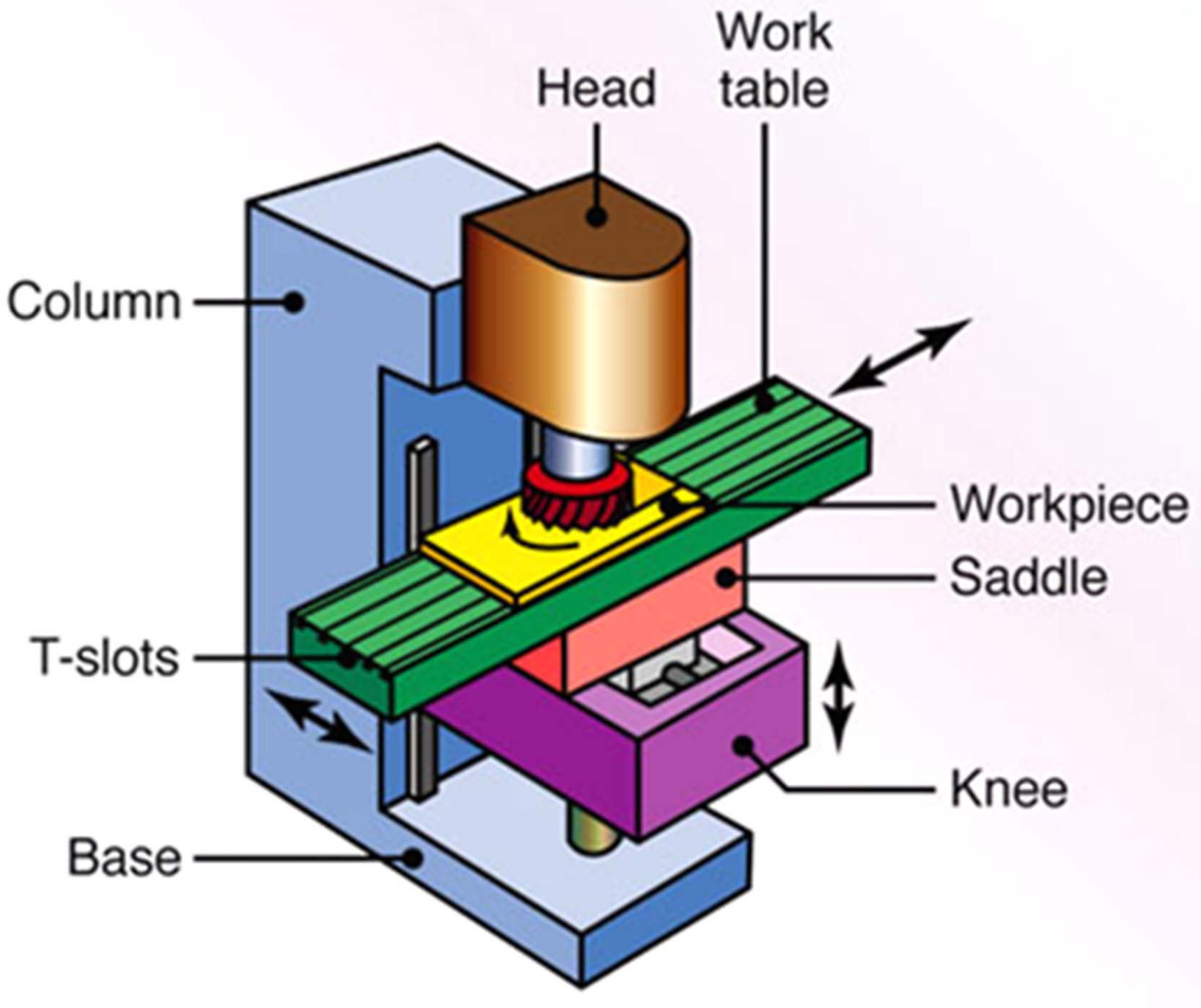
**1) Knee and column**

- Vertical
- Horizontal
- Horizontal with vertical head attachment
- Universal

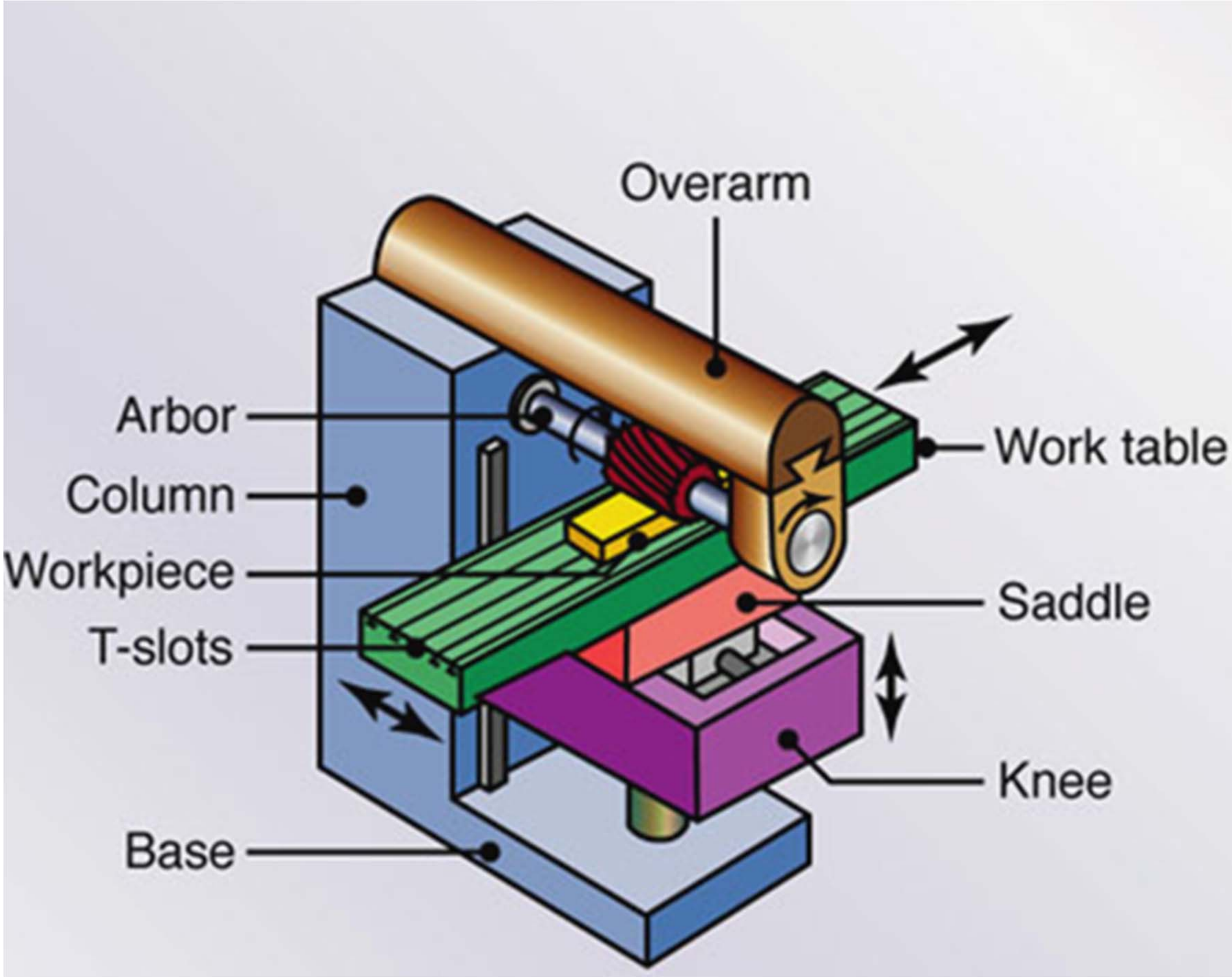
**2) Ram & Turret**

**3) Special Purpose : For production usage**

# Knee & Column: Vertical



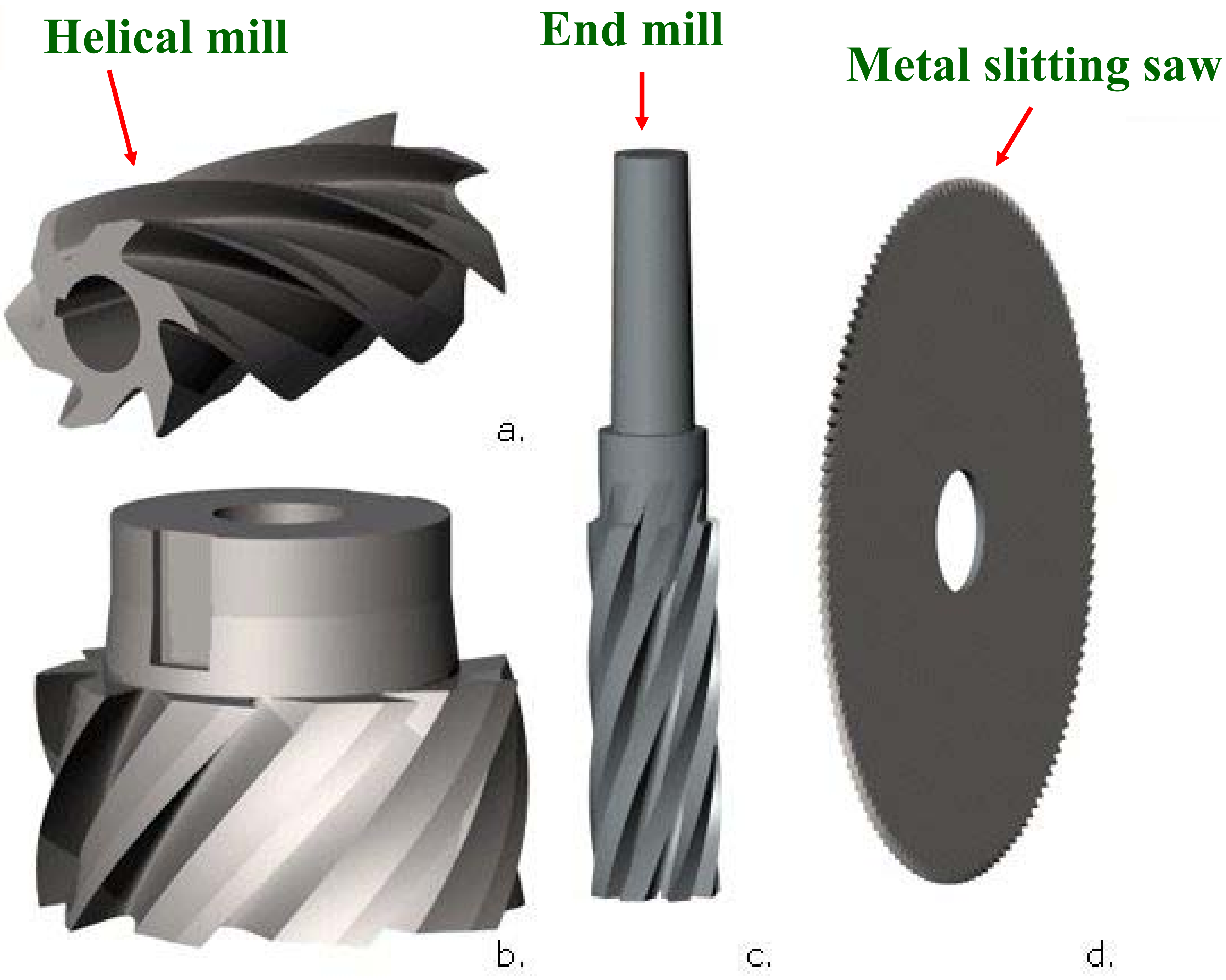
# Knee & Column: Horizontal

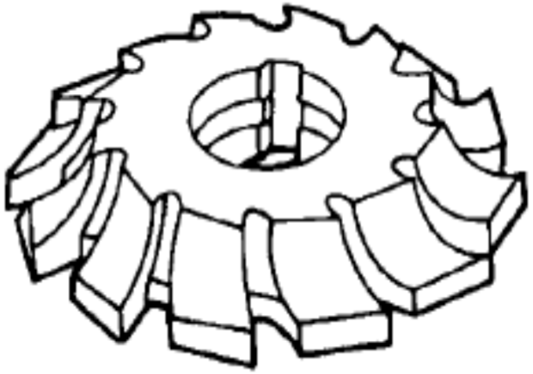


# Milling cutters

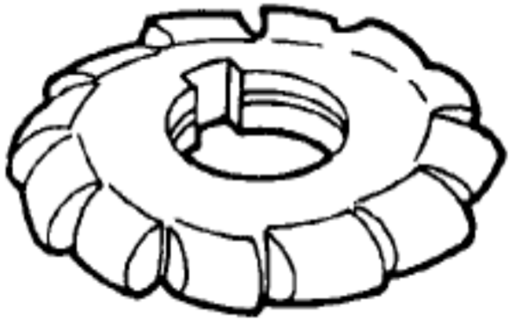
**The Basic Types:**

- End mills
- Plain milling cutter
- Face milling cutter
- T-slot cutter
- Dovetail cutter
- Form milling cutter
- Slitting cutter

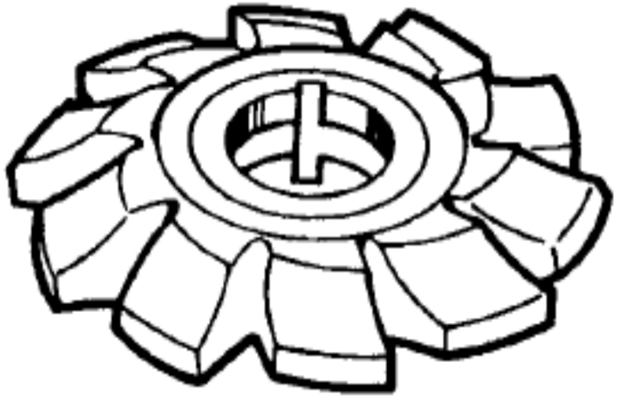

Helical mill
End mill
Metal slitting saw
a.
b.
c.
d.



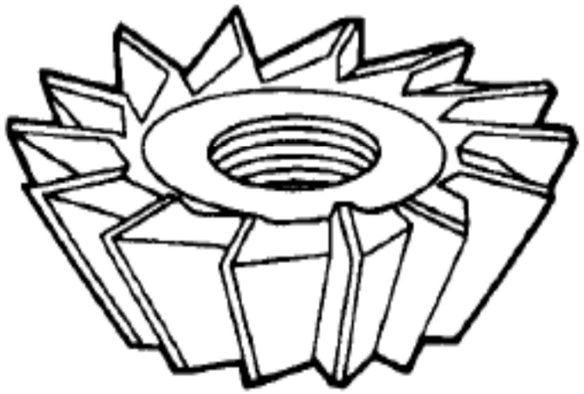
CORNER ROUNDING CUTTER



CONVEX FORMED CUTTER

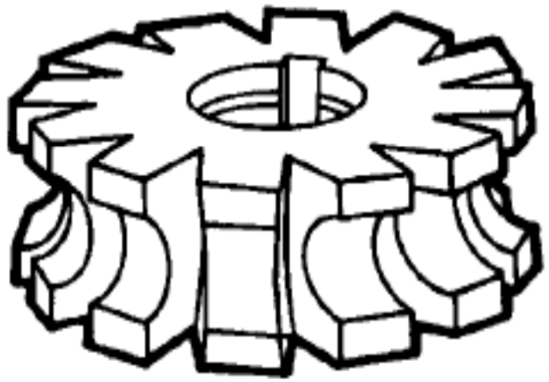


GEAR TOOTH CUTTER



SINGLE ANGLE CUTTER

DOUBLE ANGLE CUTTER



CONCAVE FORMED CUTTER

# Milling cutting mechanism

- **There are two types of the mechanisms:**

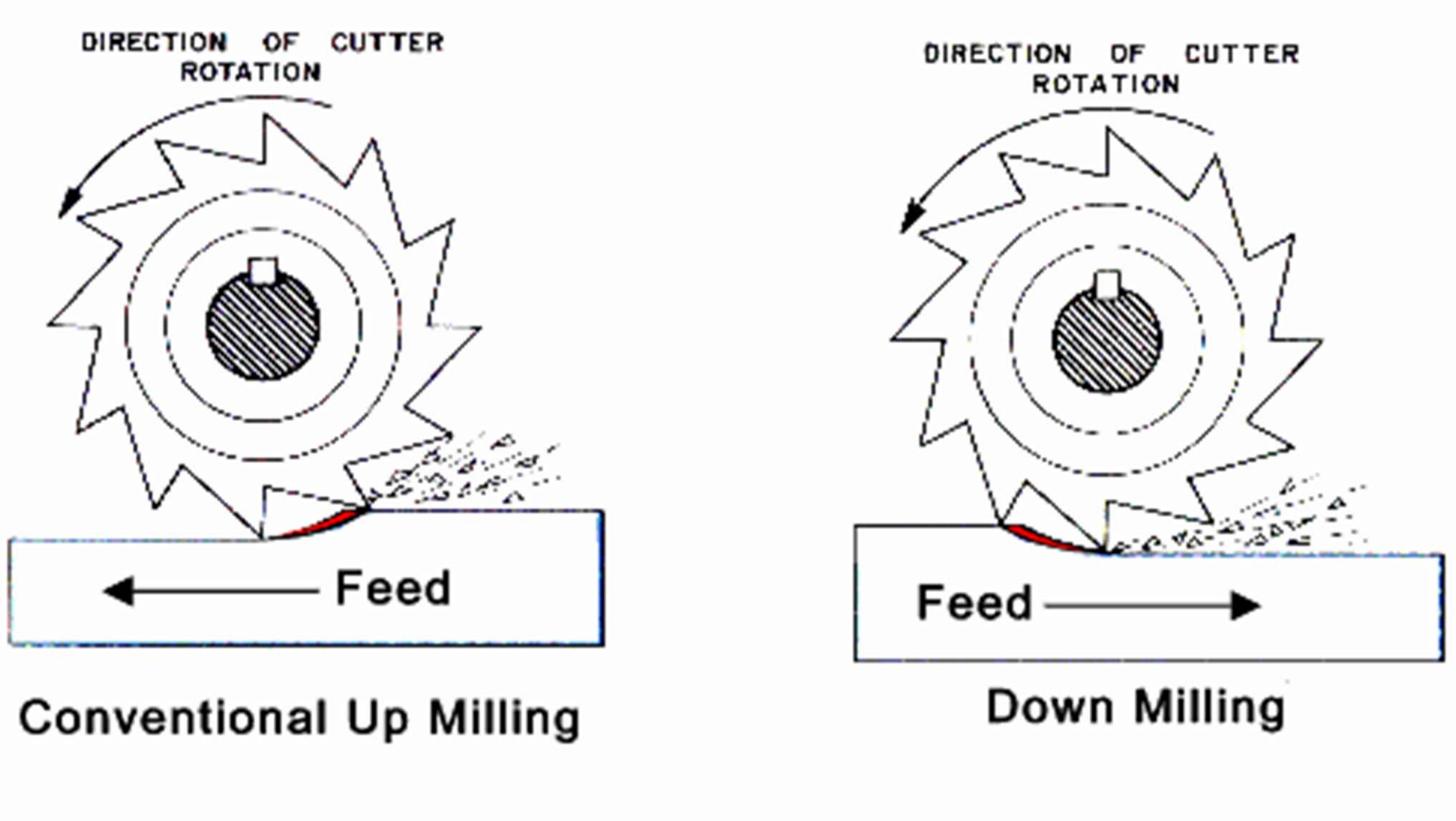
**1) Up Milling: (conventional milling)**

**In this case, the table moves towards the cutter opposing the cutter direction.**

**2) Down Milling: (climb milling)**

**When the cutter rotation is in the same direction as the motion of the work being fed, it is referred to as Down-cut or climb cut.**

# Milling cutting mechanism



# Milling cutting mechanism

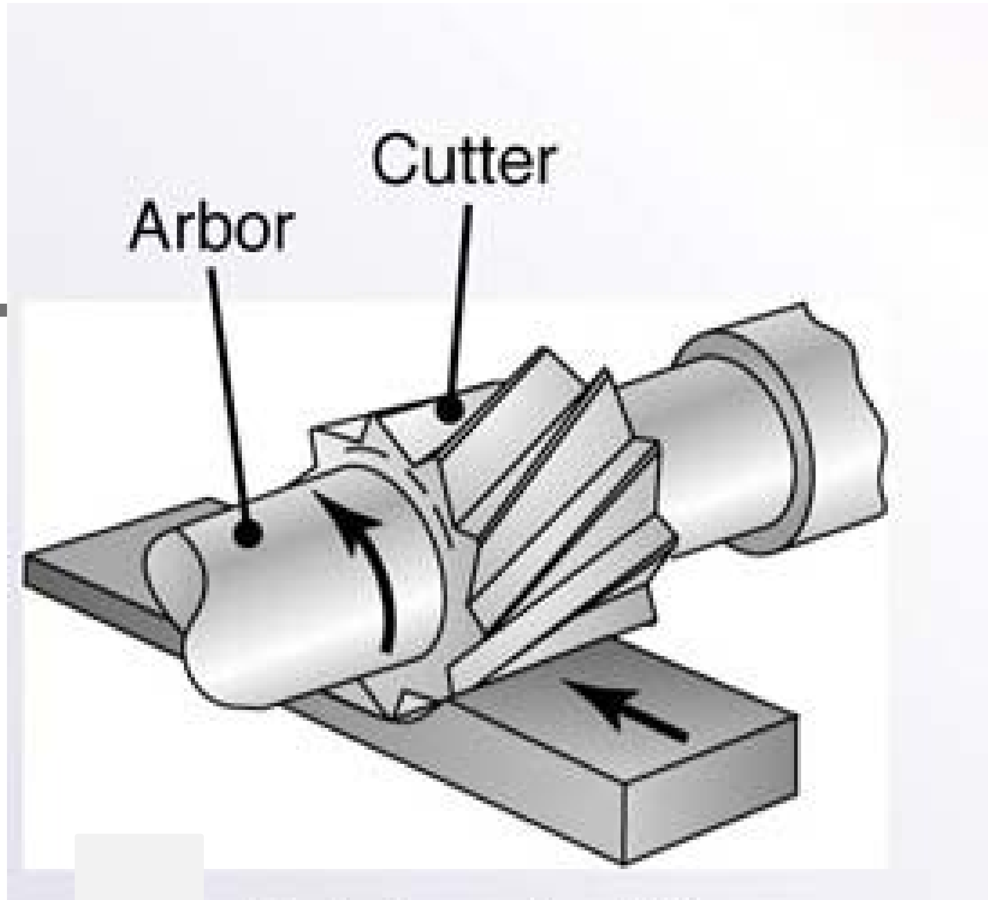
- **Down milling method advantages:**
  - The cutter forces are directed into the table, which **reduces fixture forces**, and allows thinner work pieces.
  - There is **less radial pressure** on the arbor.
  - Gives **better surface finish**.

# Milling operations

- Plain milling
- Face milling
- Straddle milling
- Concave milling
- Corner rounding milling
- Convex milling
- Slot milling
- Slit milling
- T-slot milling, dove tail slot milling
- Gang milling
- Gear milling

# Milling operations

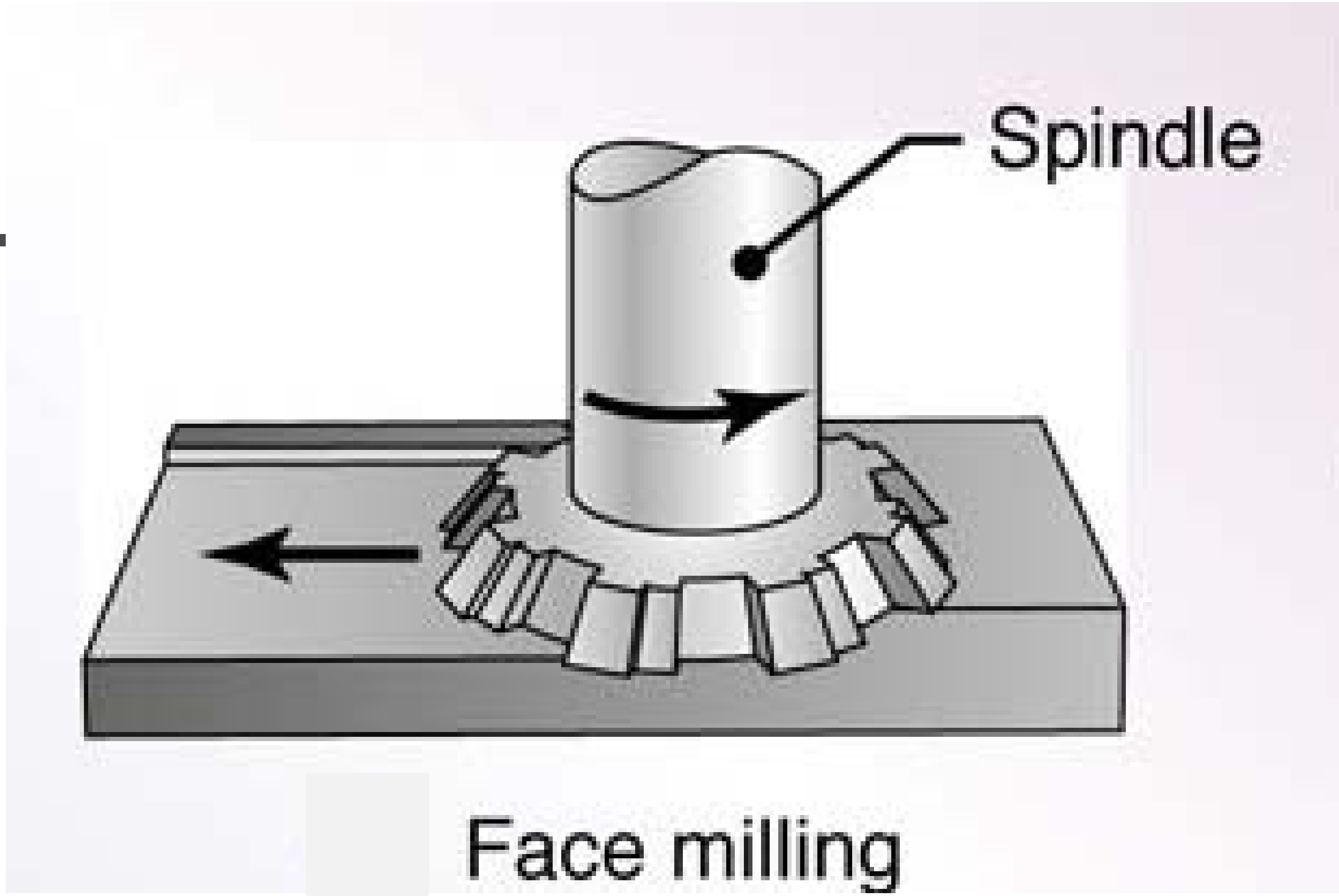
## Peripheral (plain) milling




Peripheral milling

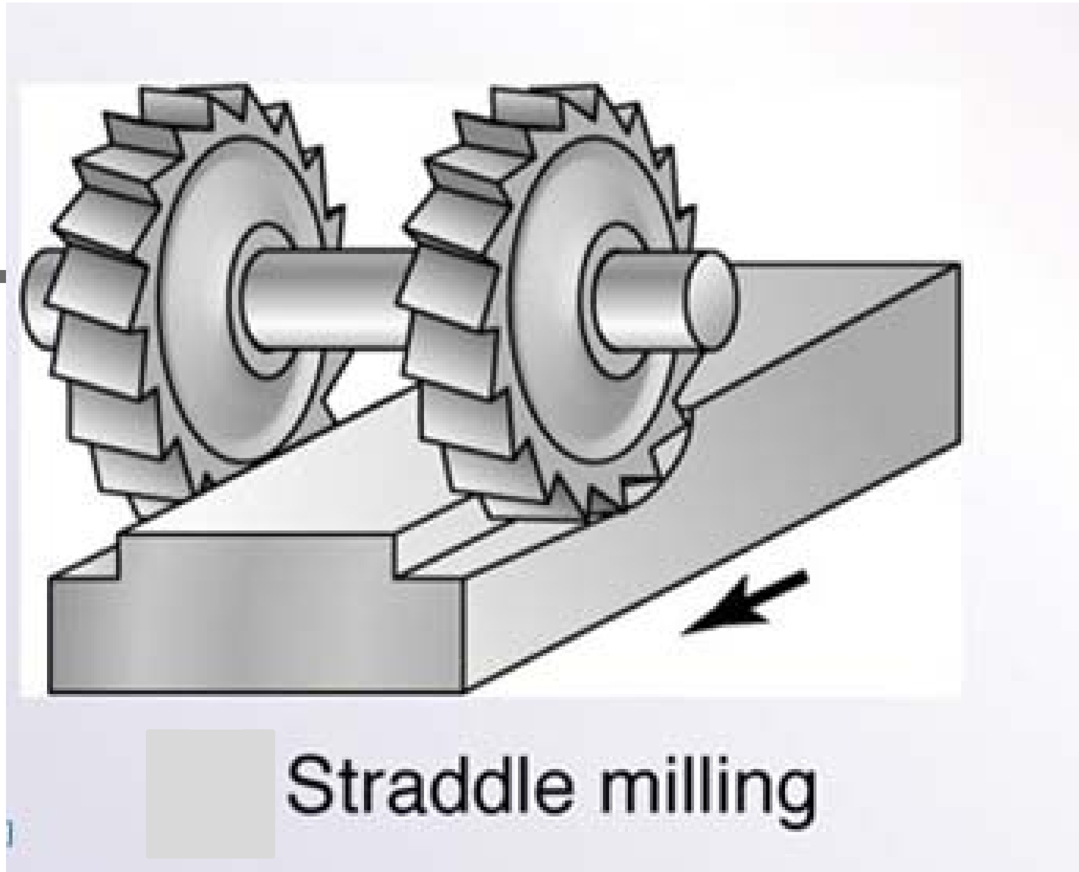
# Milling operations

**Face milling**



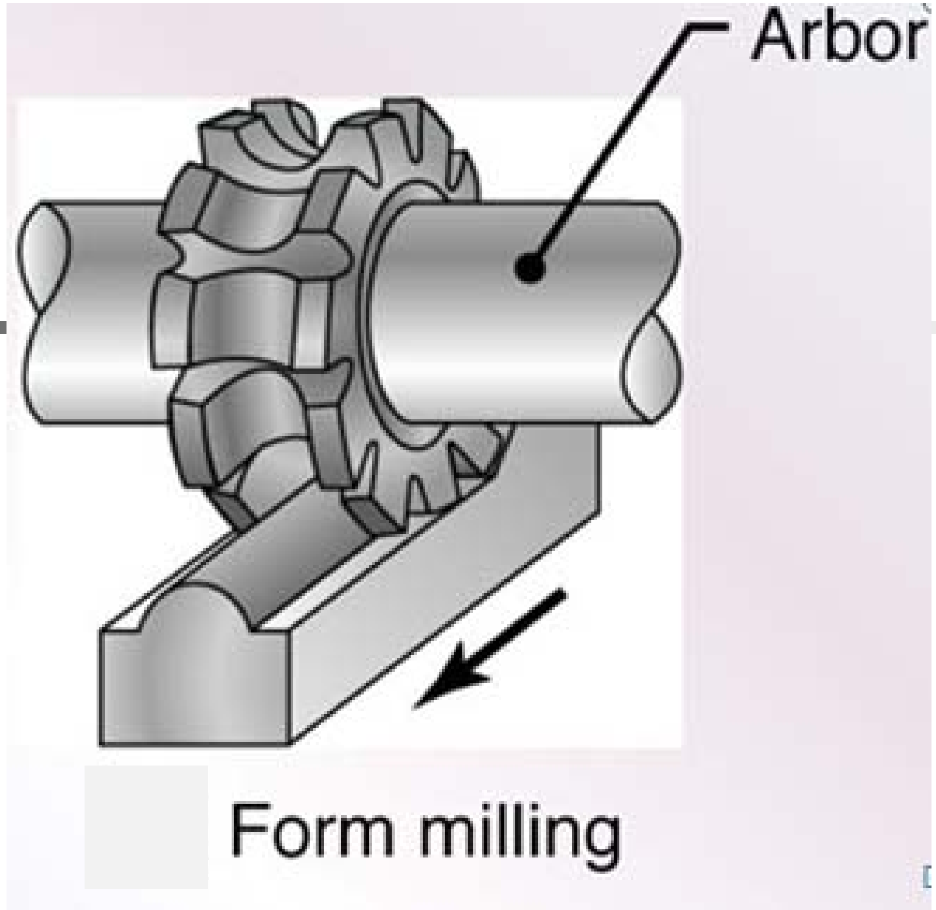
# Milling operations

Straddle milling



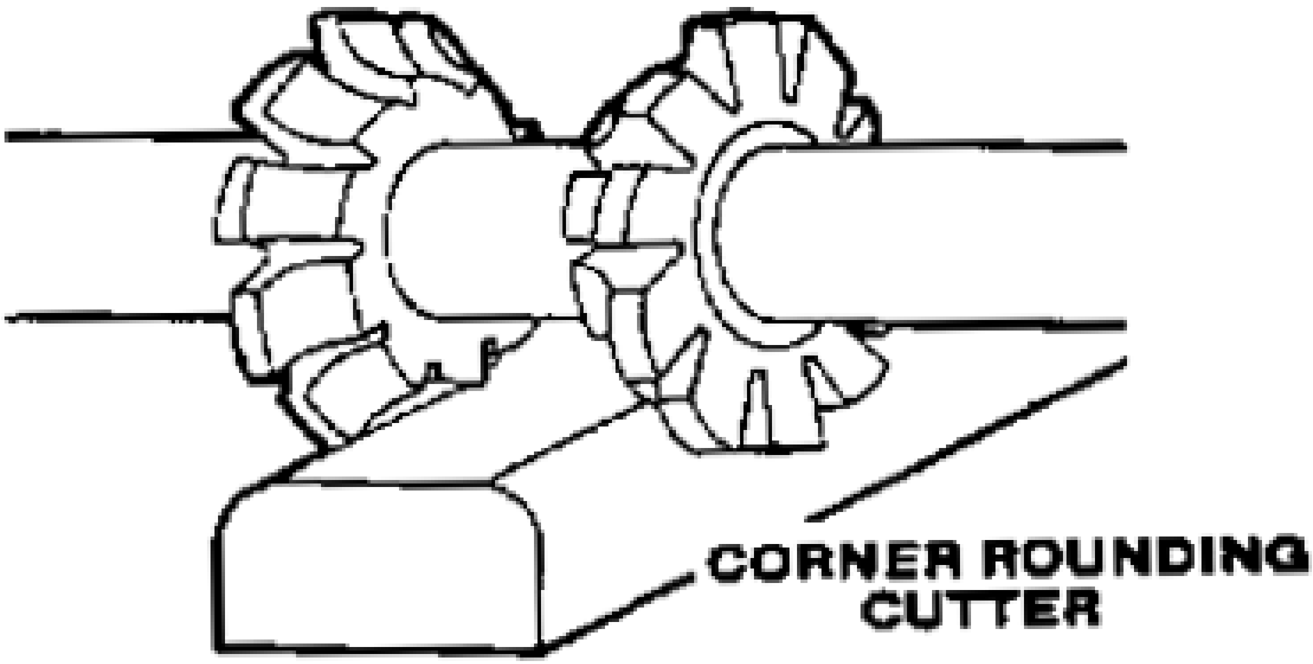
# Milling operations

**Concave milling**



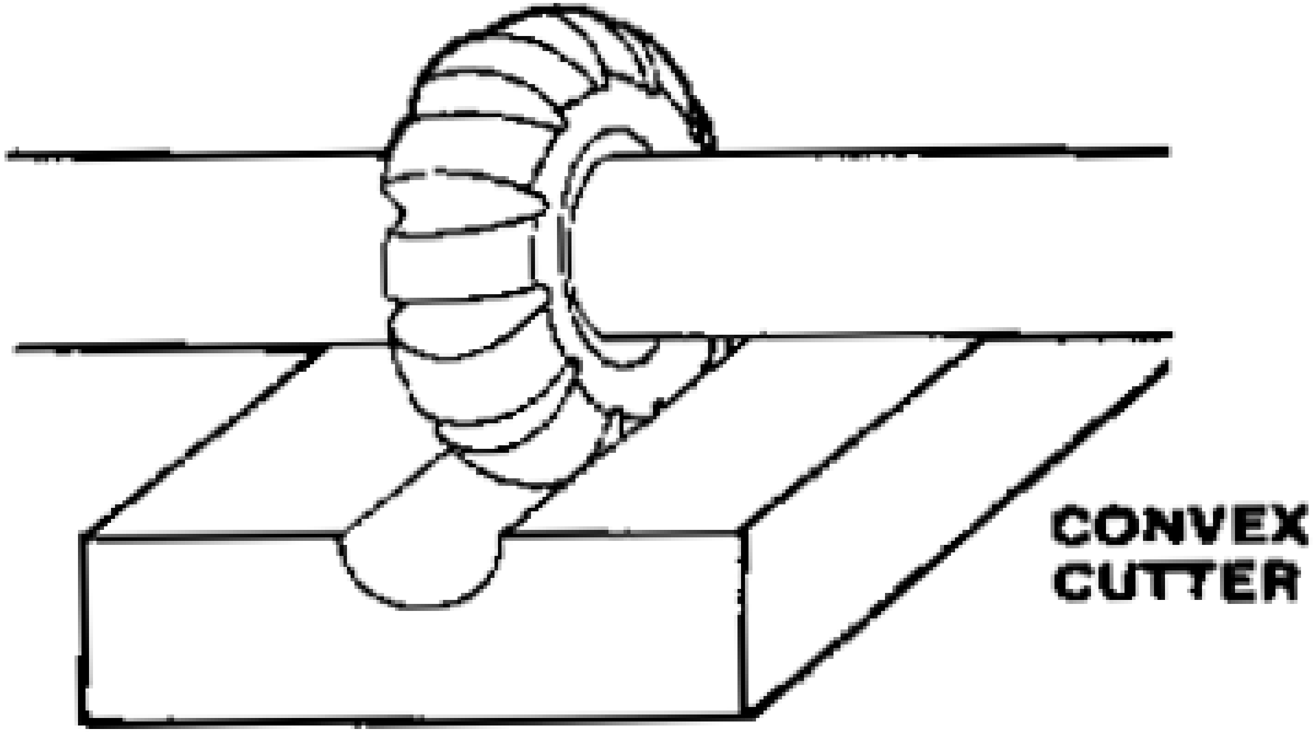
# Milling operations

## Corner rounding milling



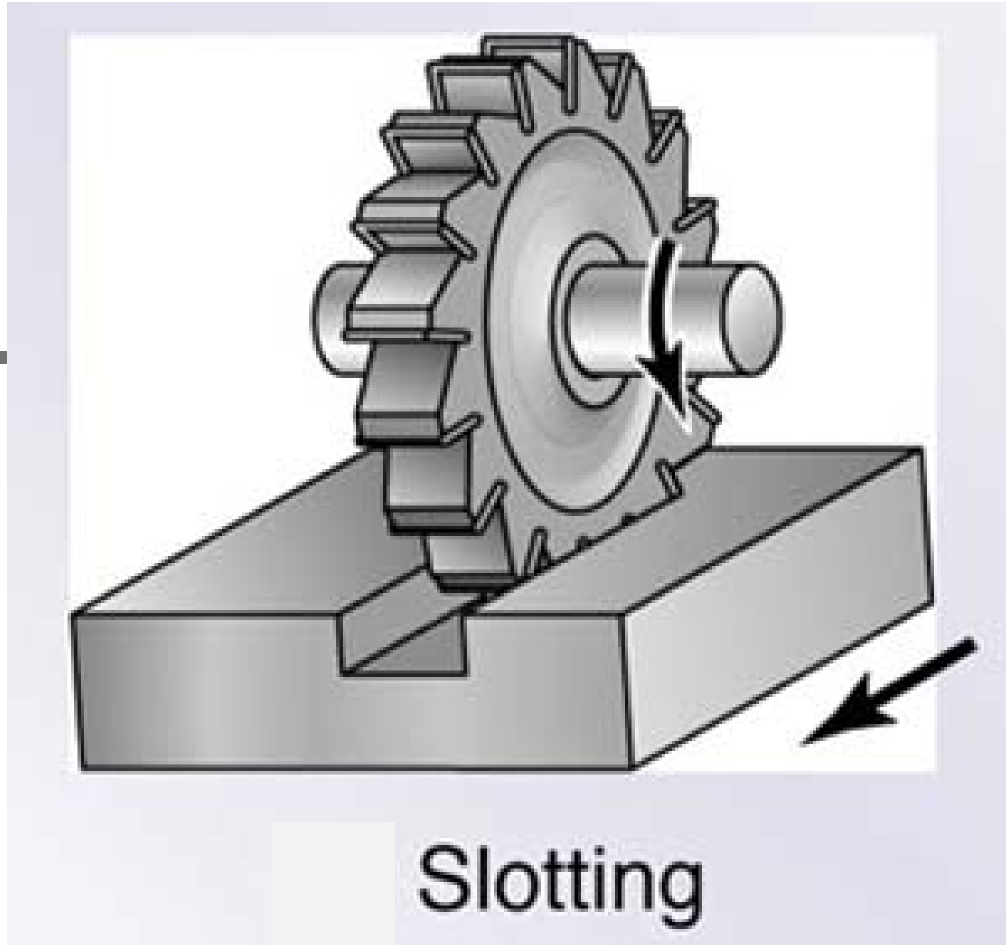
# Milling operations

Convex milling



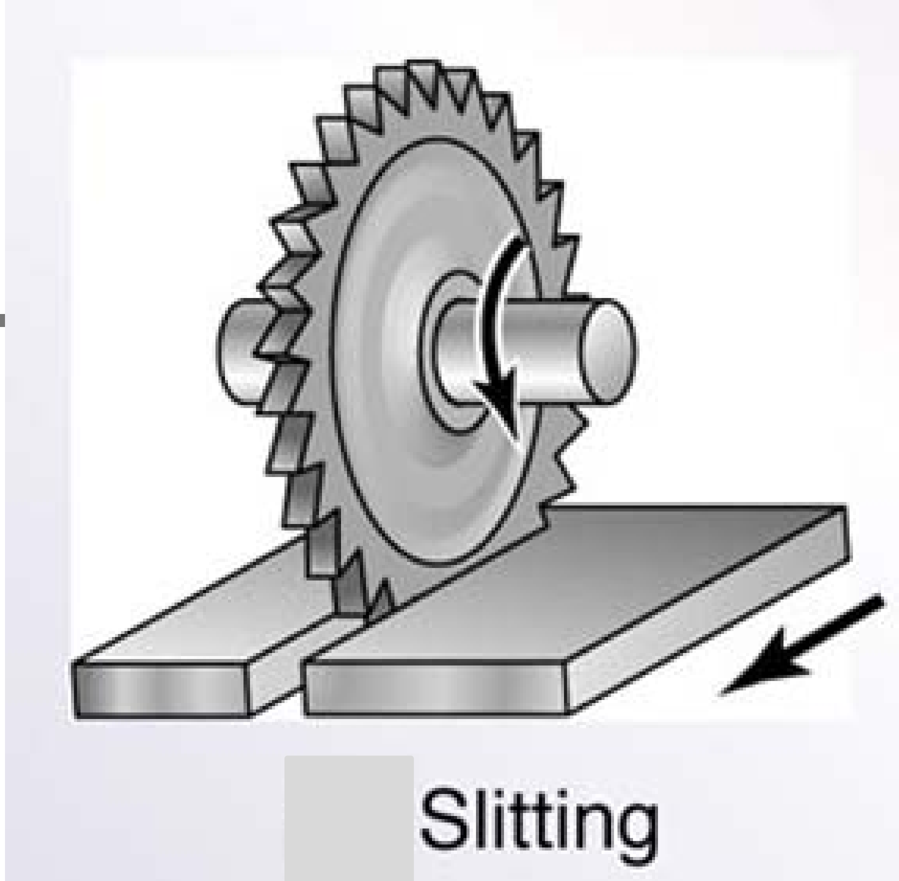
# Milling operations

Slotting milling



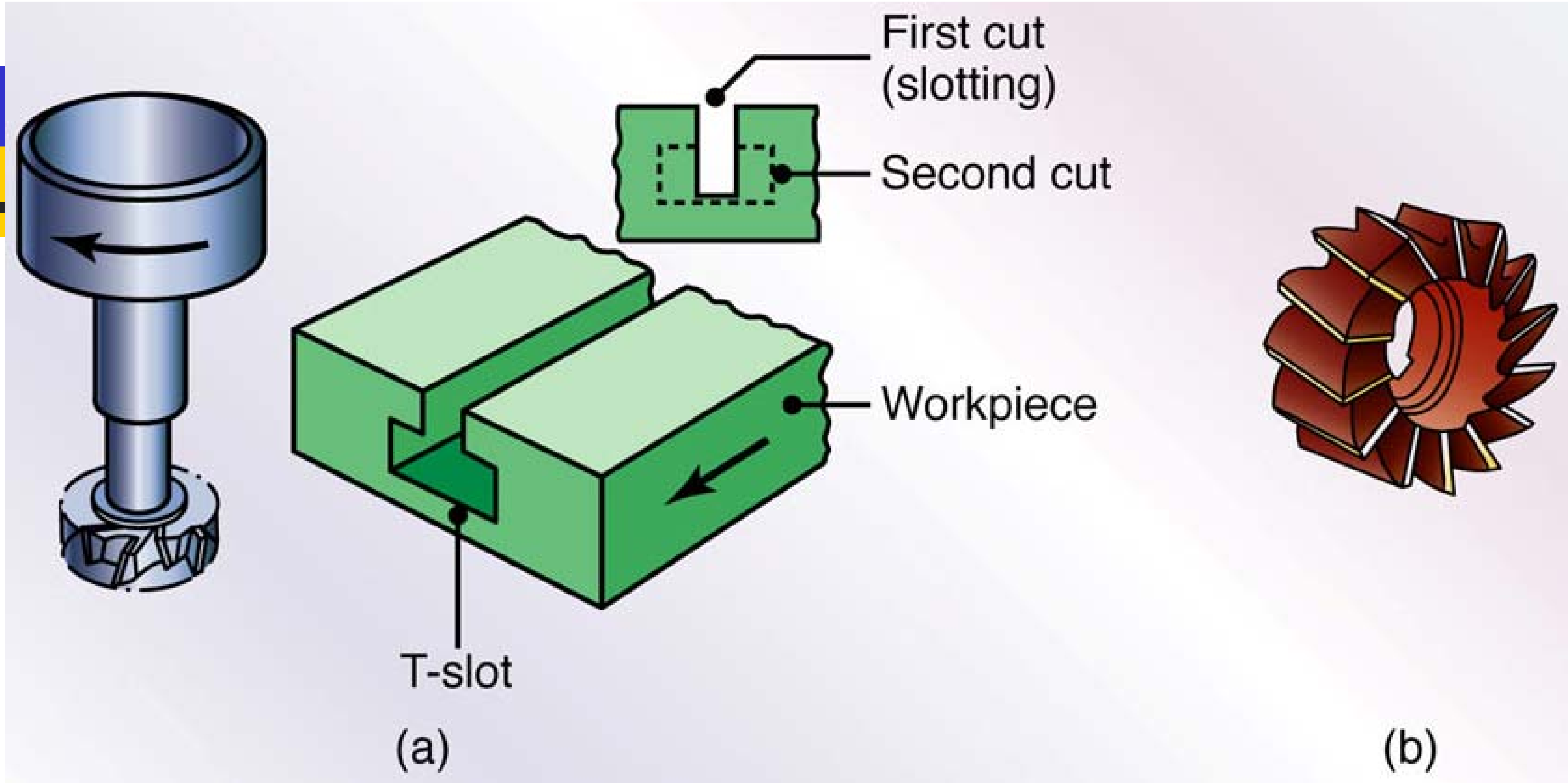
# Milling operations

**Slitting milling**



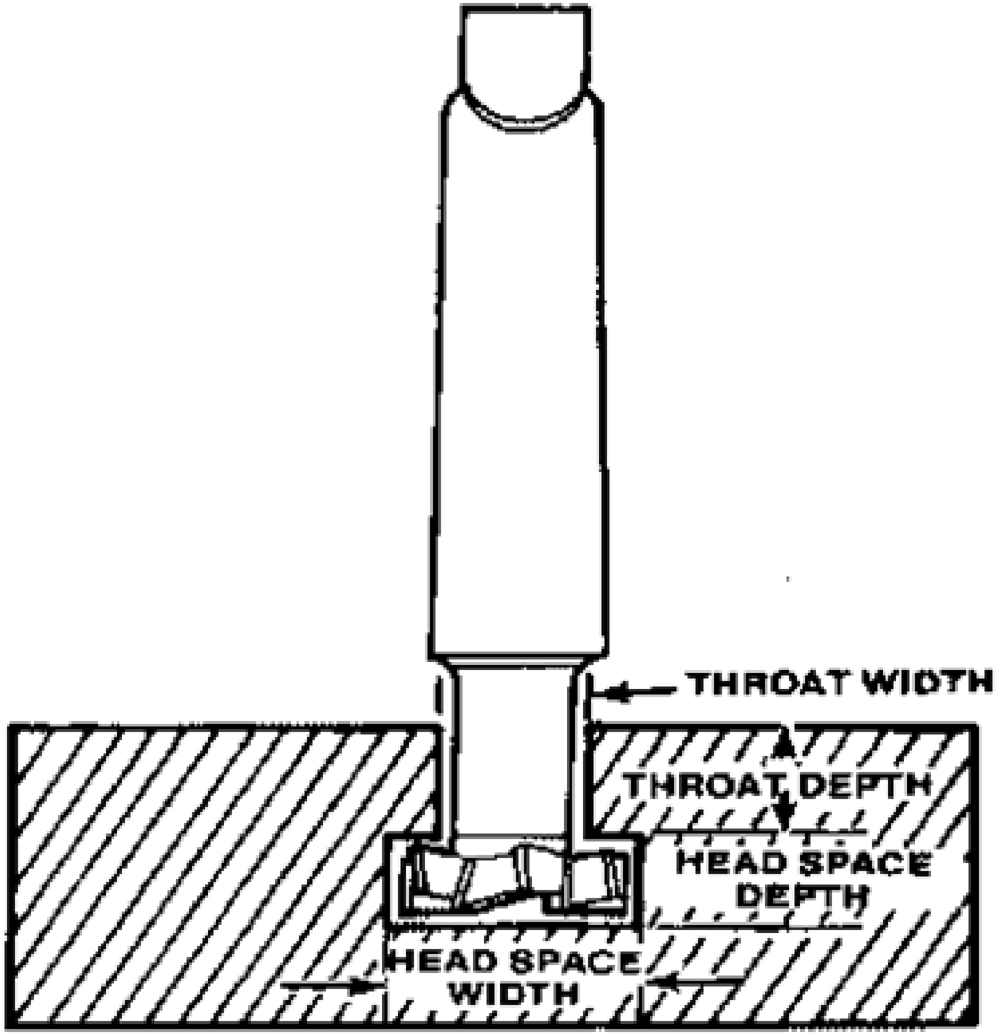
# Milling operations

## T-slot milling



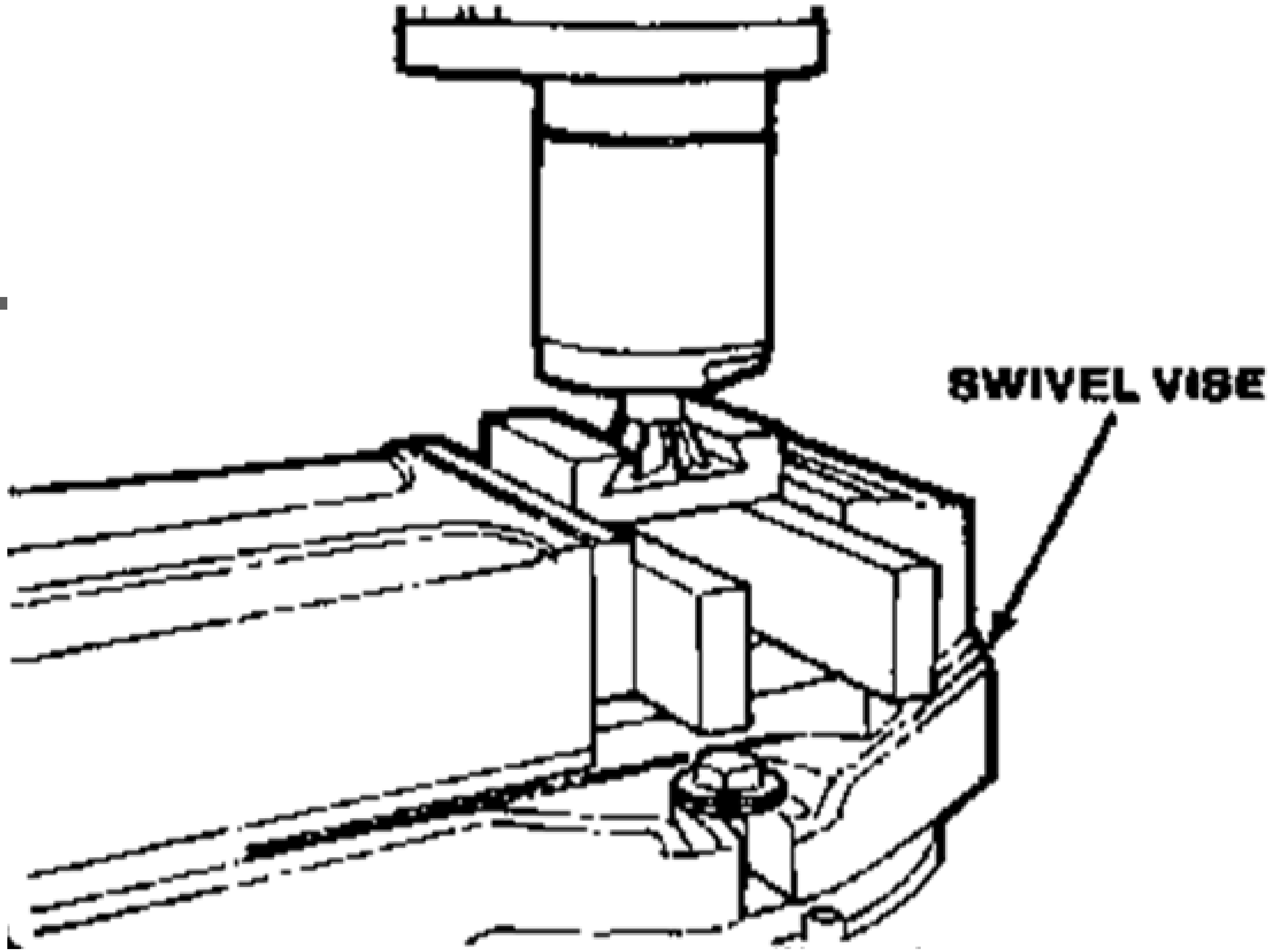
# Milling operations

## T-slot milling



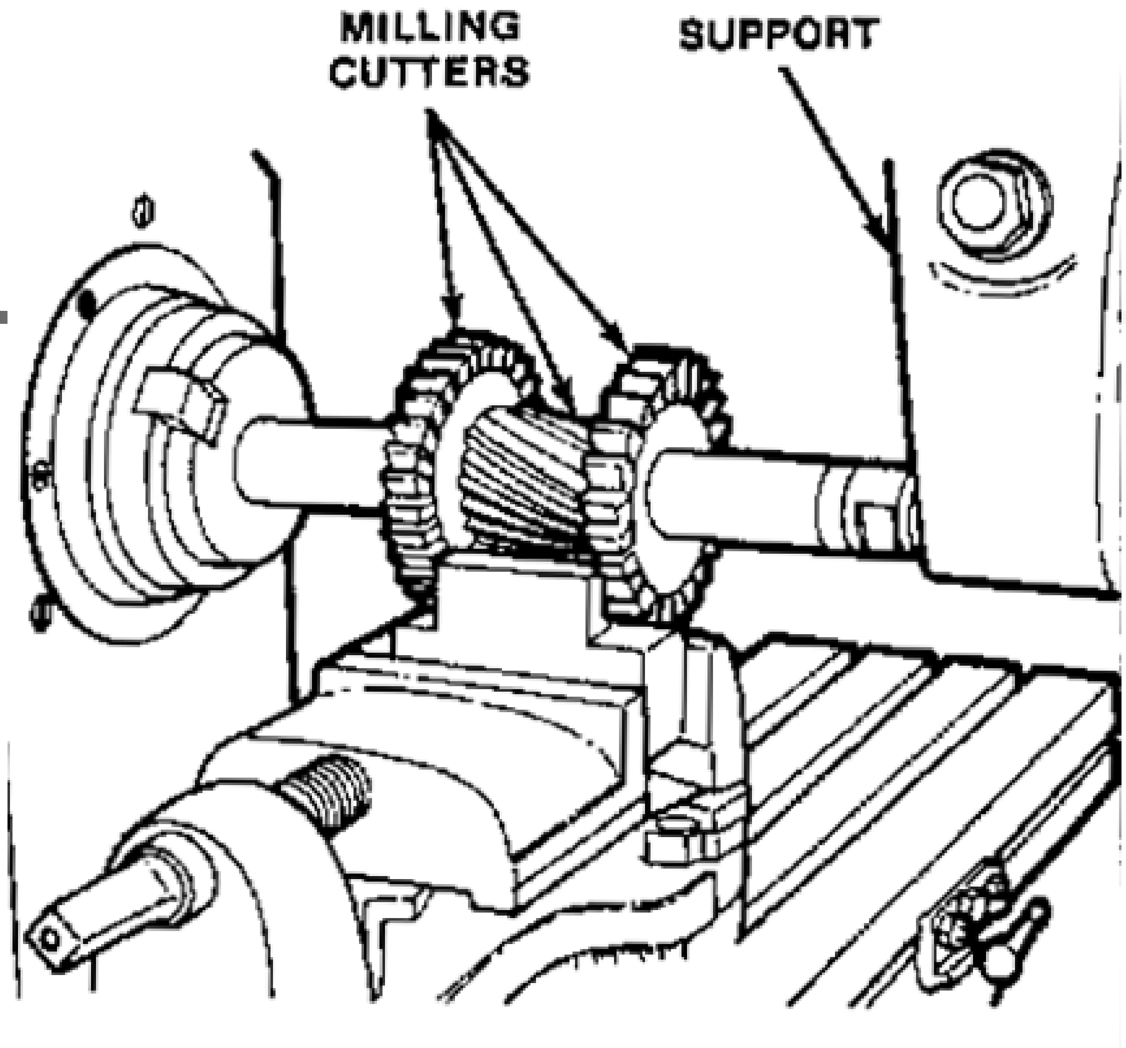
# Milling operations

Dove tail milling



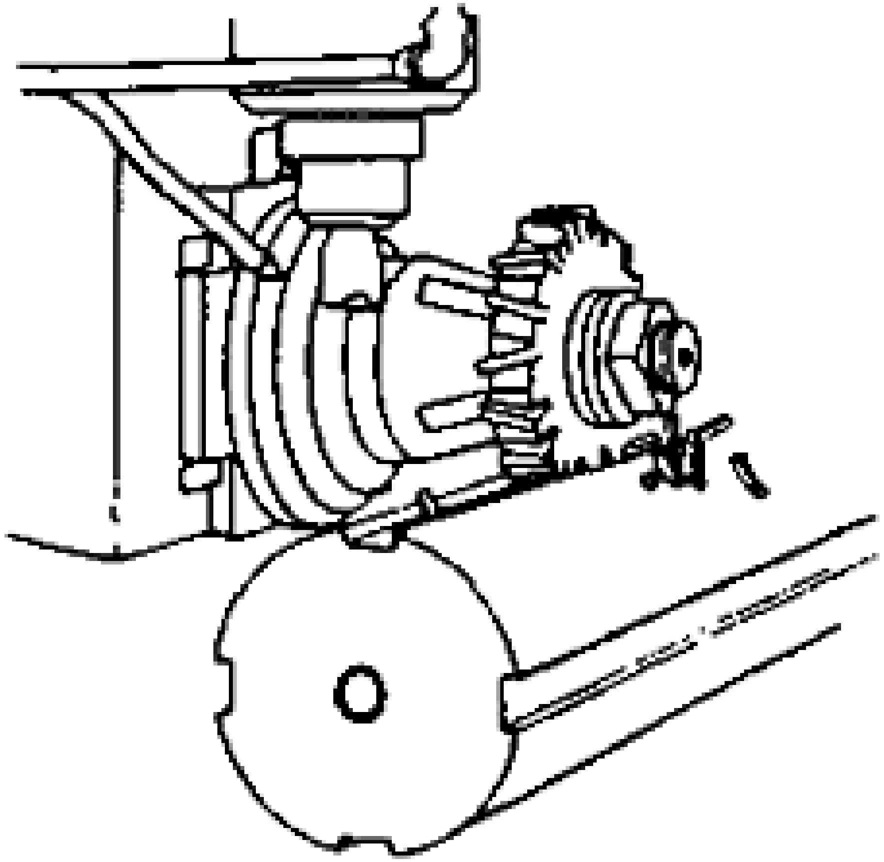
# Milling operations

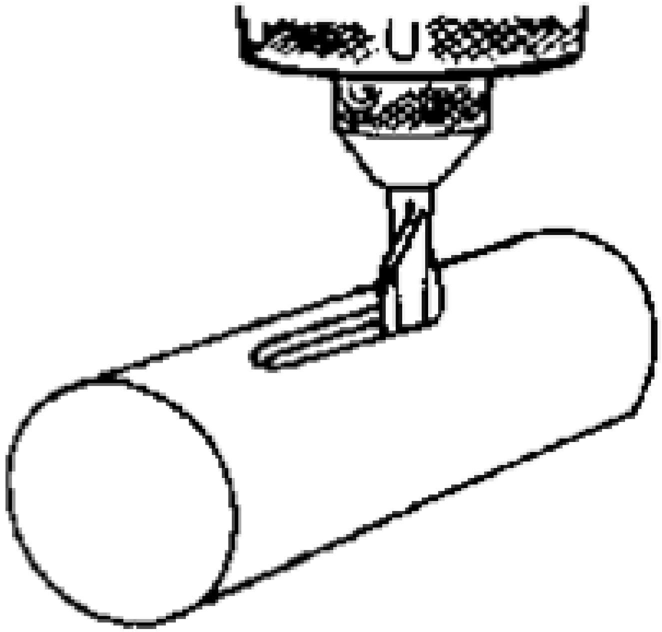
**Gang milling**



# Milling operations

## Key slot milling

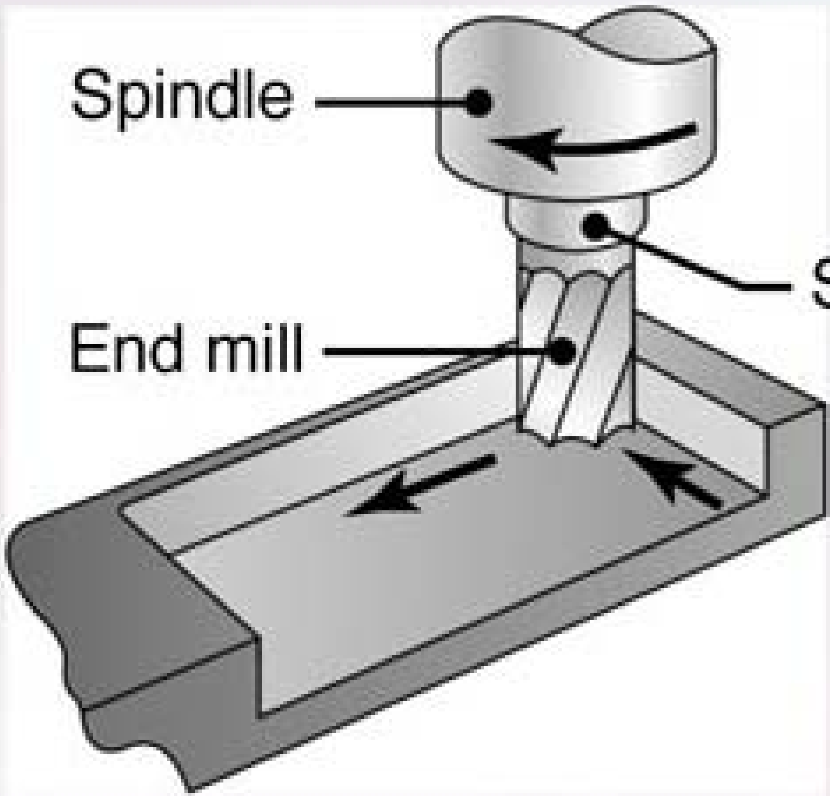




AFTER CENTERING AND SETTING DEPTH, KEY IS MILLED TO REQUIRED LENGTH.

# Milling operations

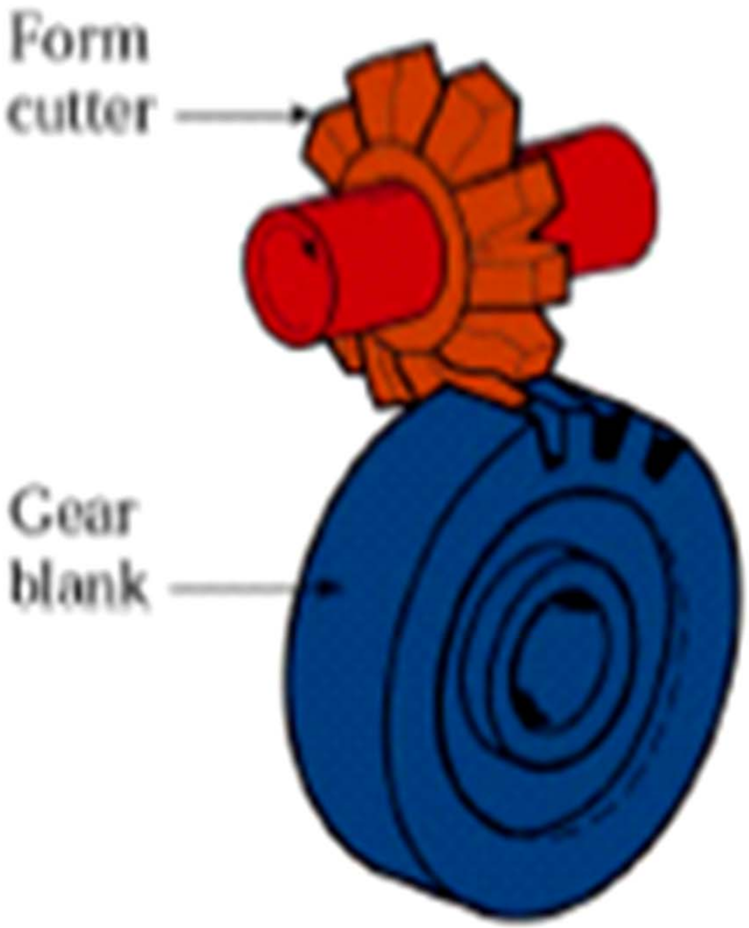
## End milling

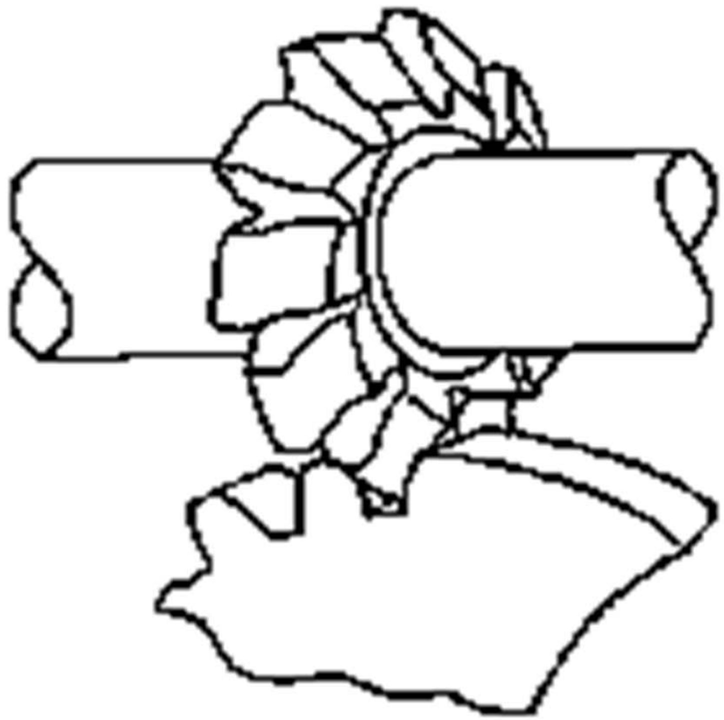



End milling

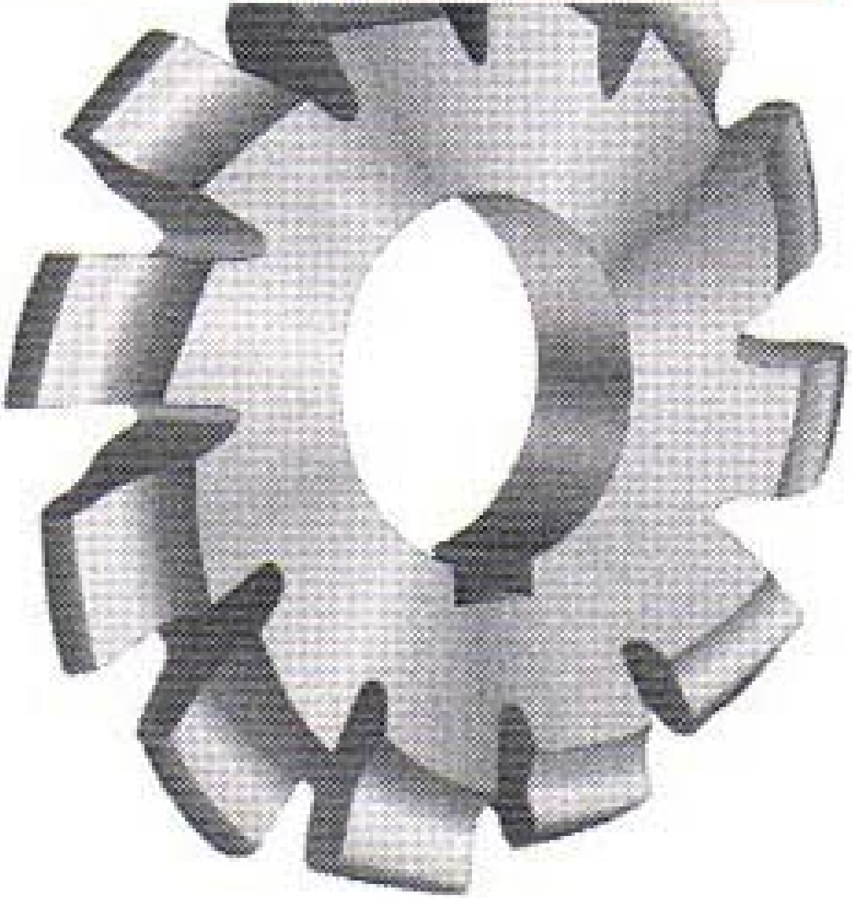
# Milling operations

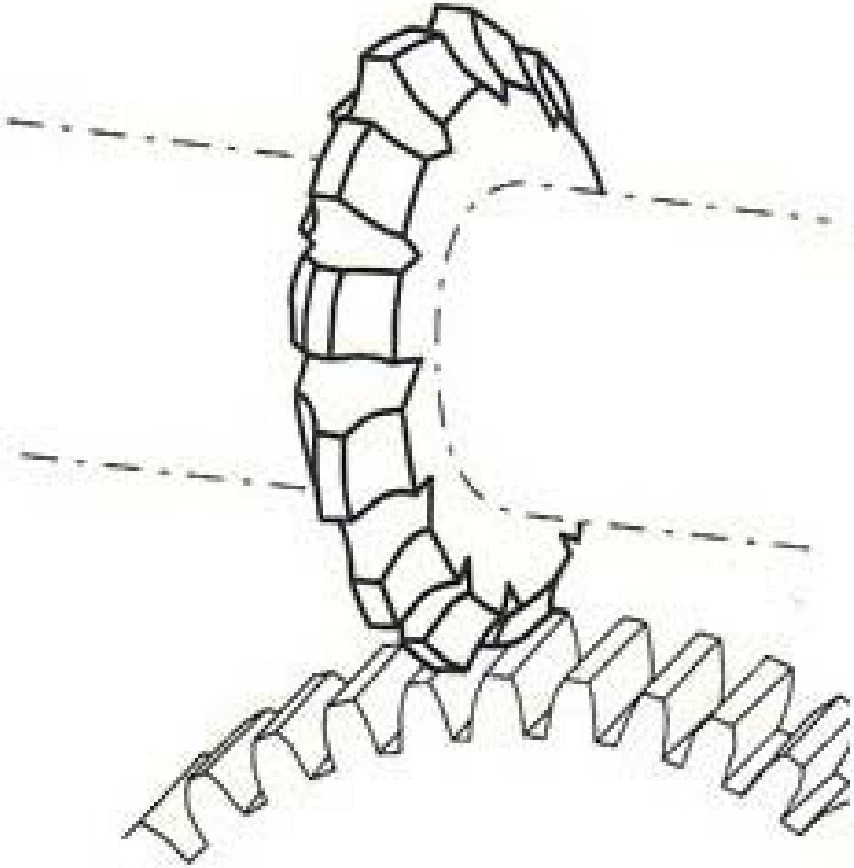
**Gear milling**



MAXWELL





# Ways of tool fixation

- **The tool can be fixed by:**

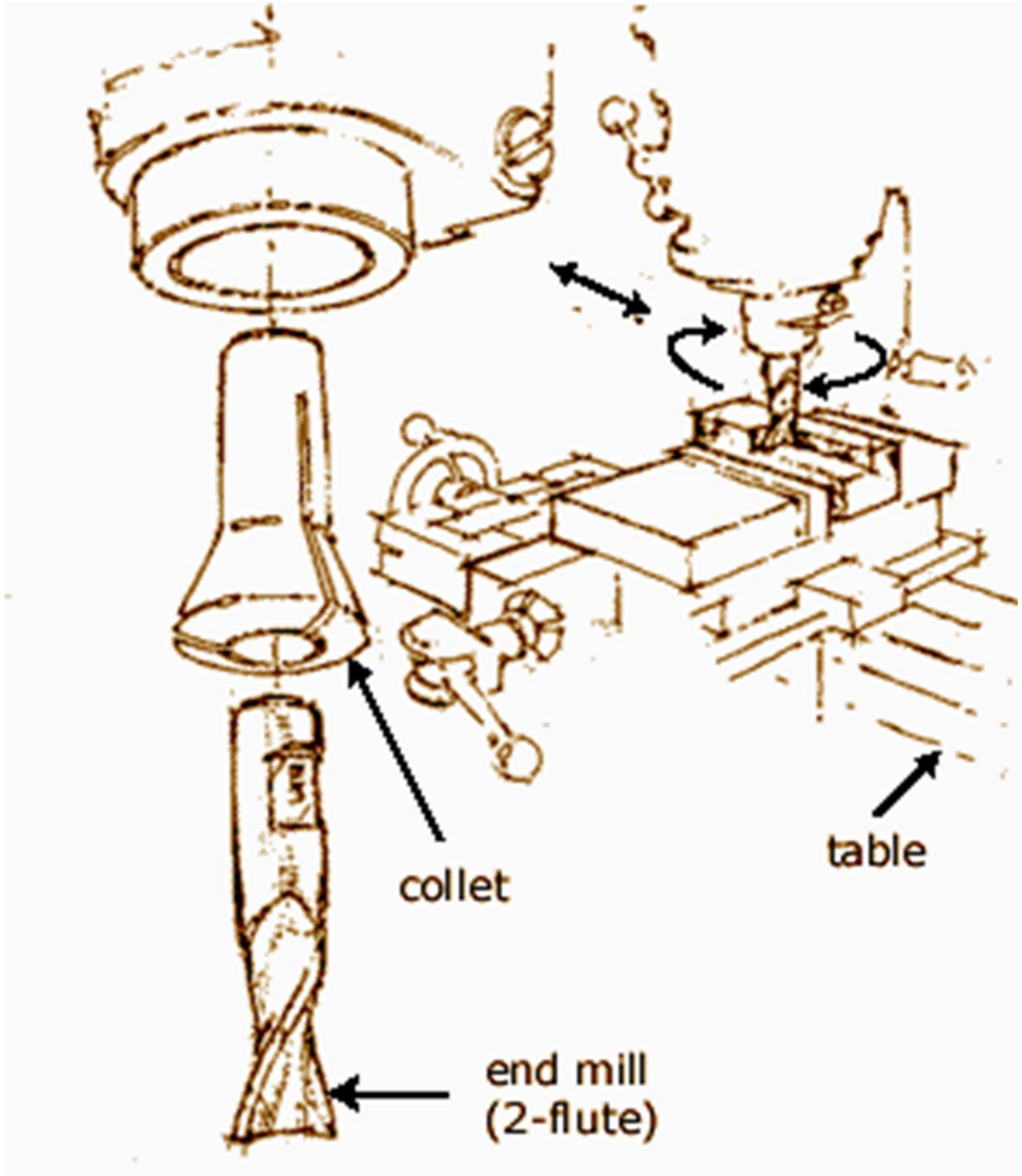
  **- Using collets**

  **- Using chucks**

  **- Using arbor**

# Ways of tool fixation

- **Collets**



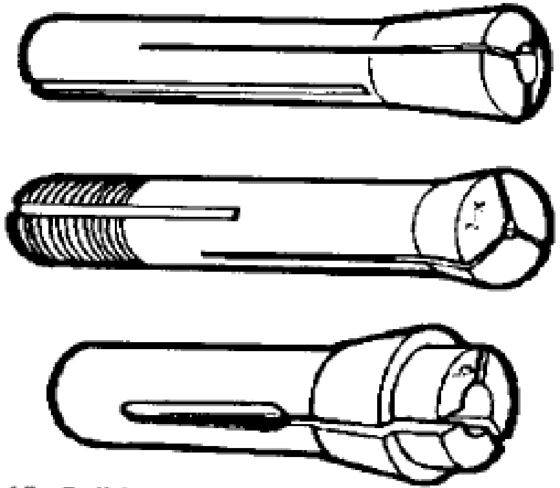
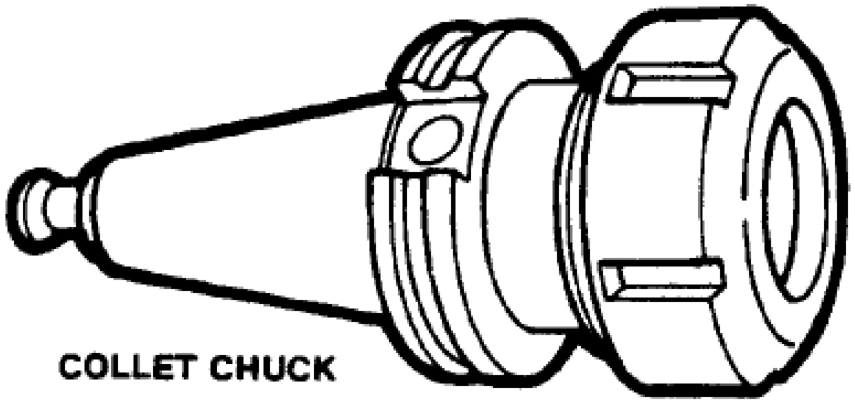
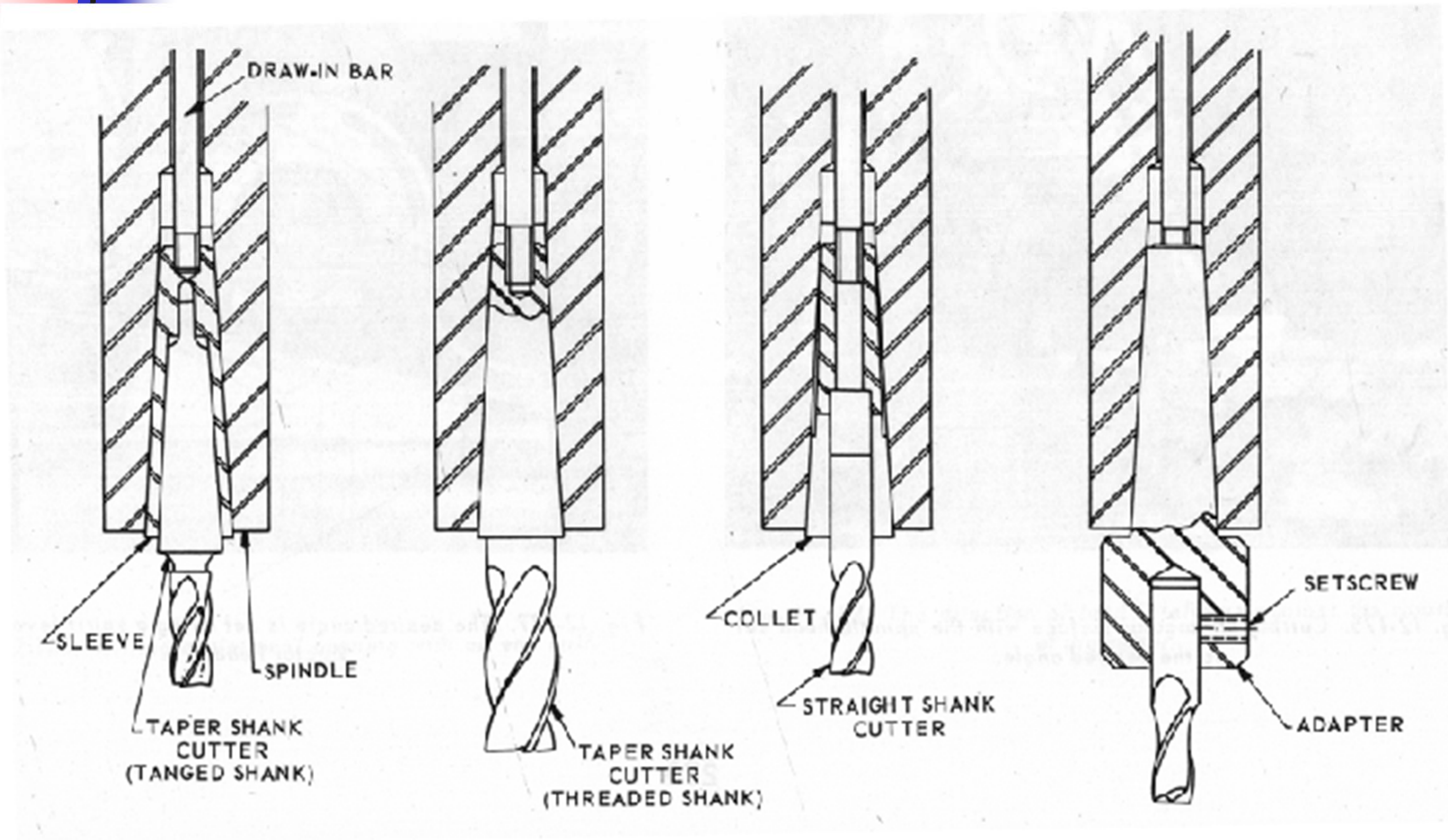


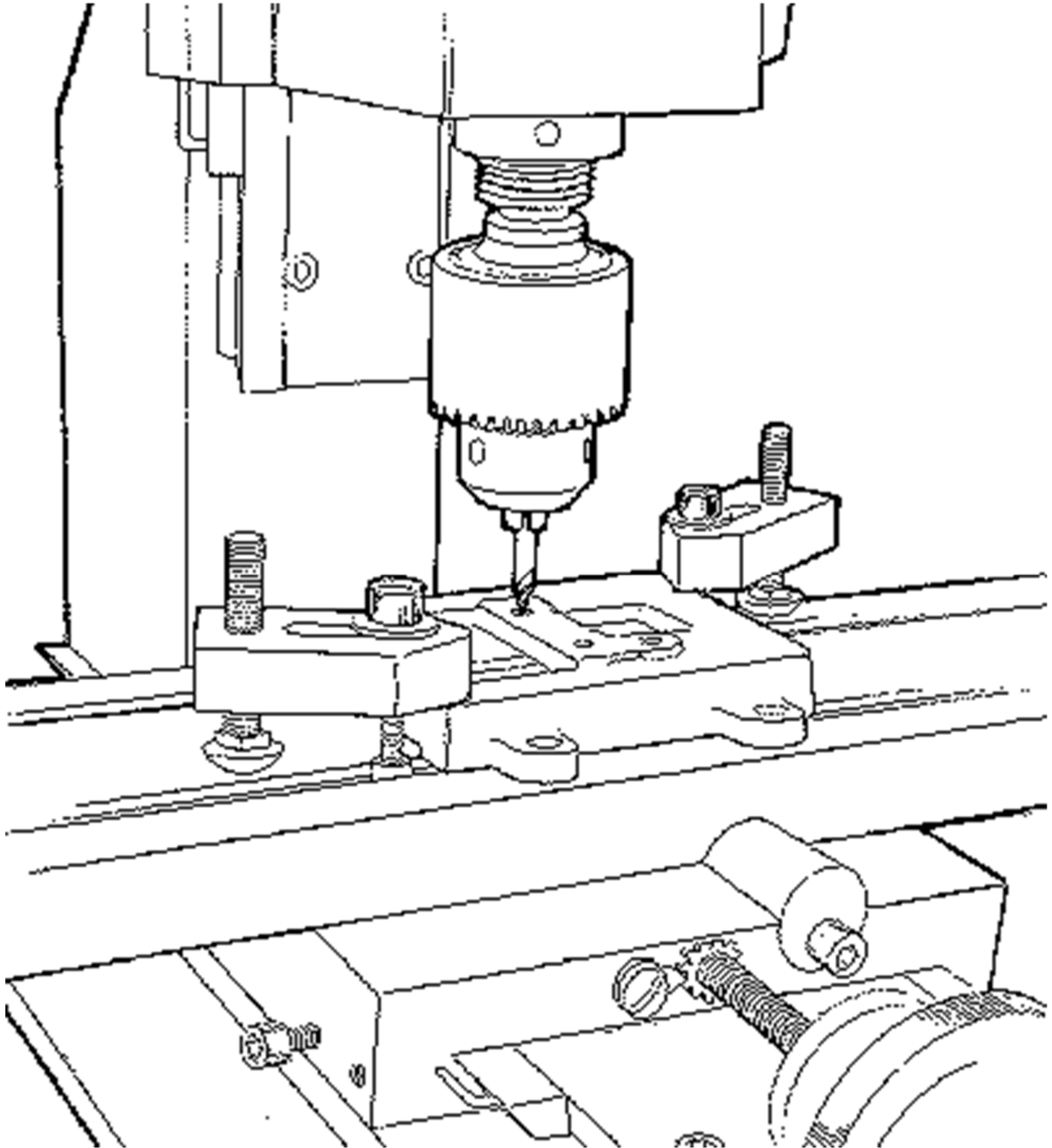
Figure 8-16. Solid and spring collets.




DRAW-IN BAR
SLEEVE
SPINDLE
TAPER SHANK
CUTTER
(TANGED SHANK)
TAPER SHANK
CUTTER
(THREADED SHANK)
COLLET
STRAIGHT SHANK
CUTTER
SETSCREW
ADAPTER

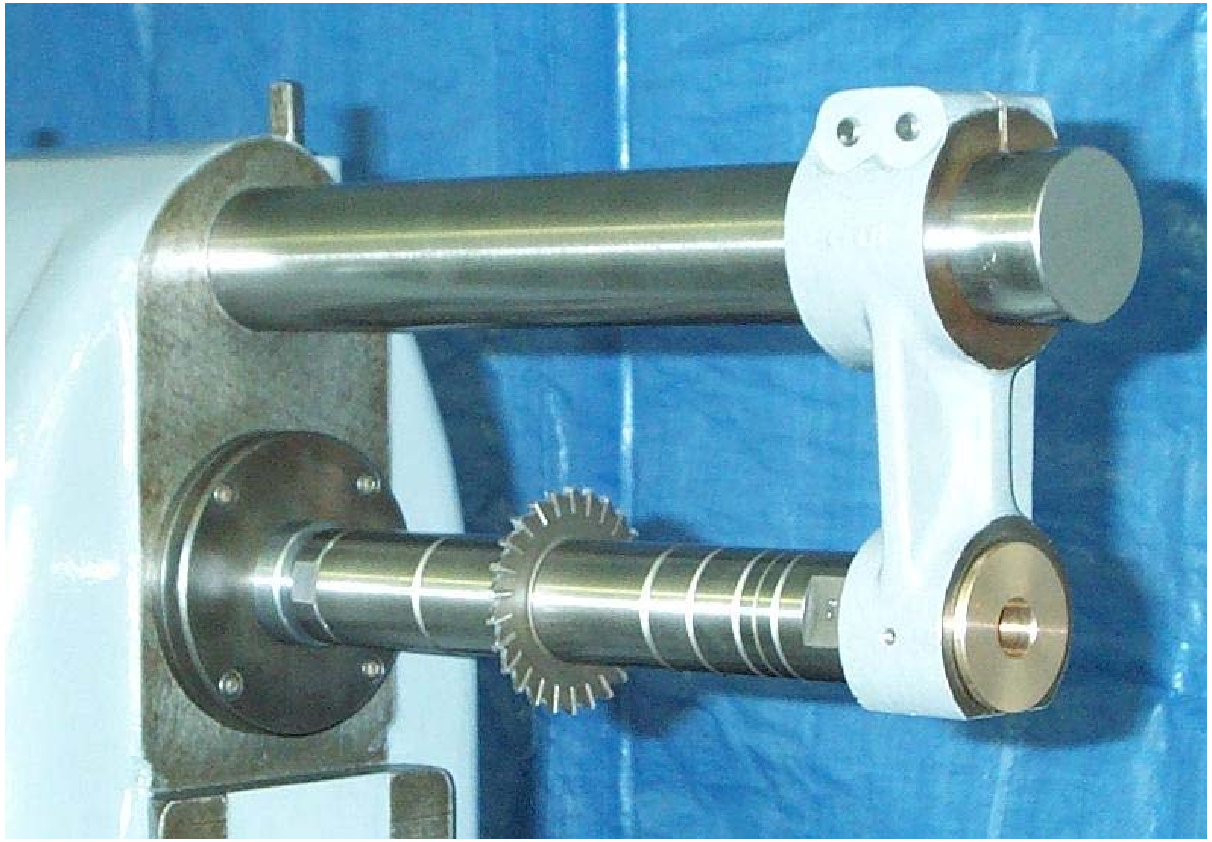
# Ways of tool fixation

- Chucks

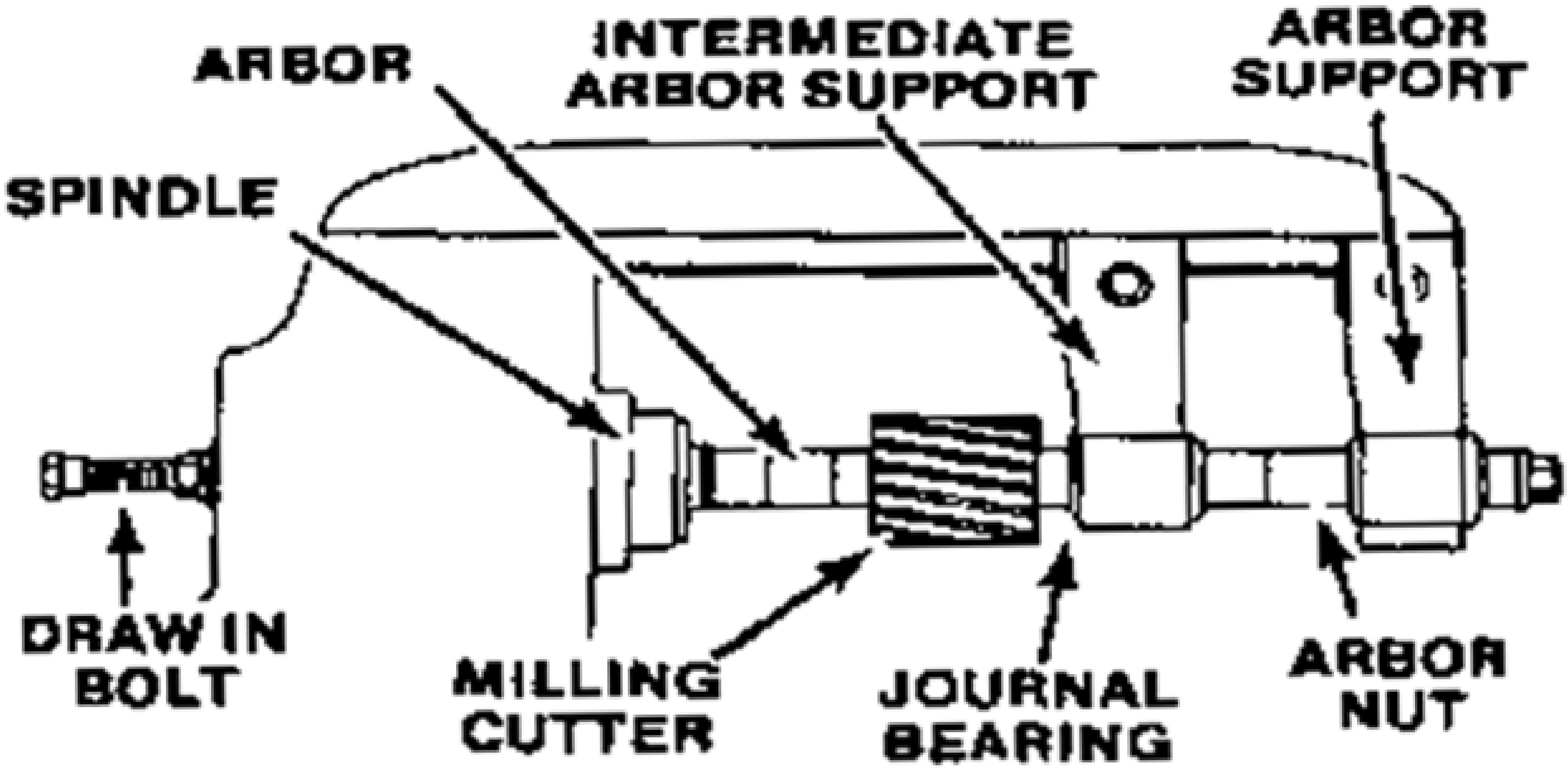


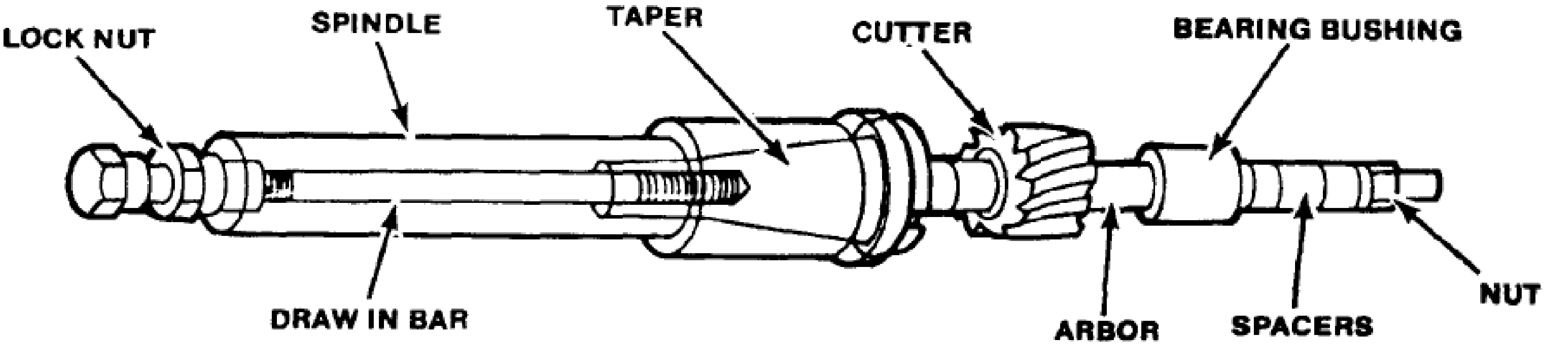
# Ways of tool fixation

- **Arbor**



# Arbor milling





A draw in bar holds the arbor firmly in the spindle.

# Arbor milling

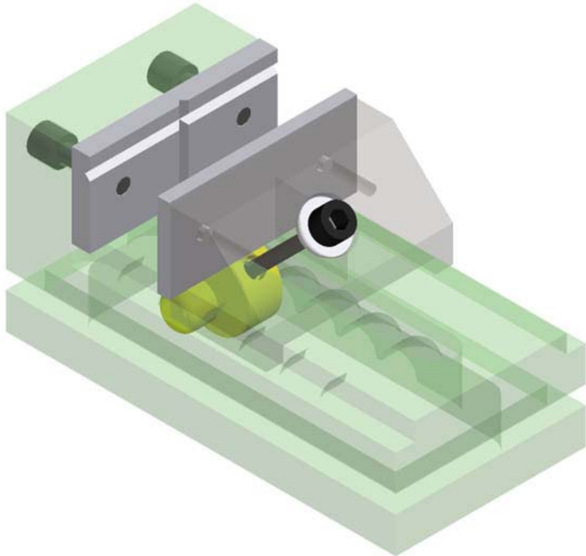
- **The advantages of arbor milling are:**
  - The cutter is held more rigidly on the spindle nose.
  - There is less variation in the arbor torque.
  - Lower power requirements.
  - Flatter work surface finish.

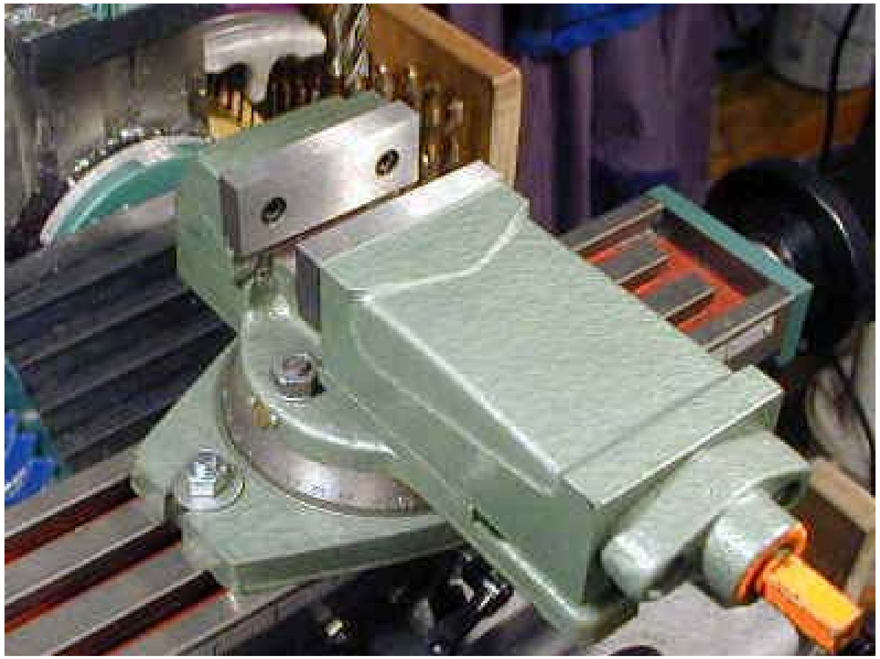
# Ways of workpiece fixation

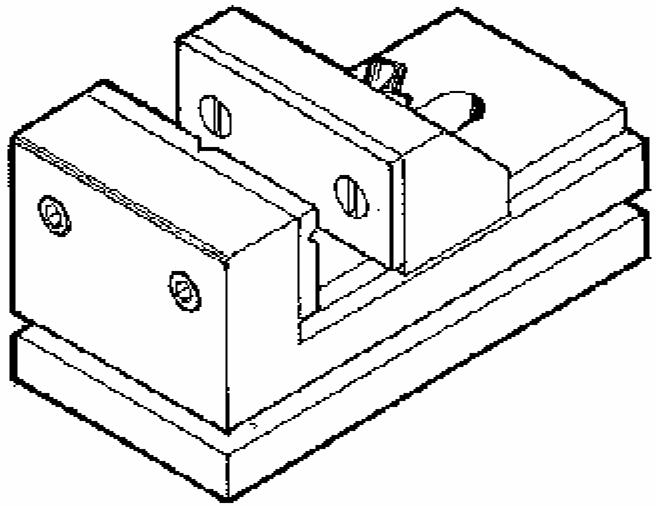
- Using vice
- Directly on the milling table
- Between two centers
- Using indexing box
- Rotary table

# Ways of workpiece fixation

- Vice = Vise

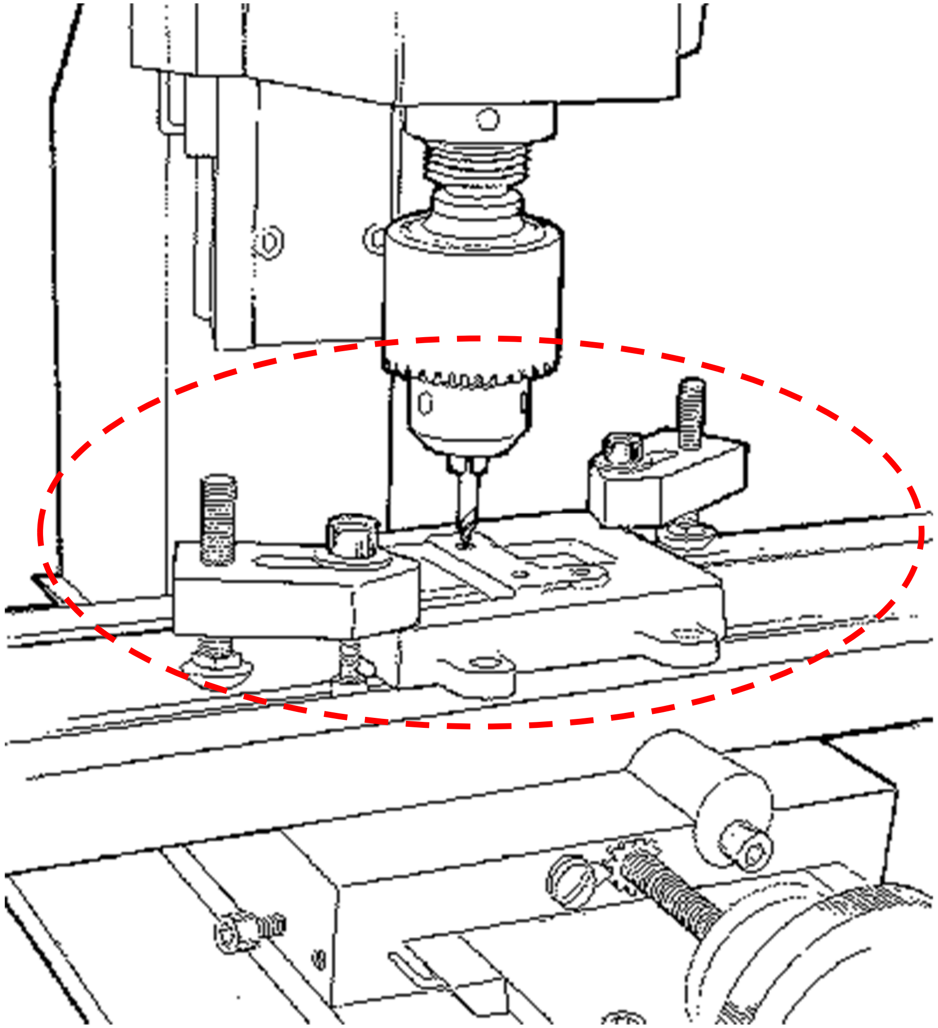






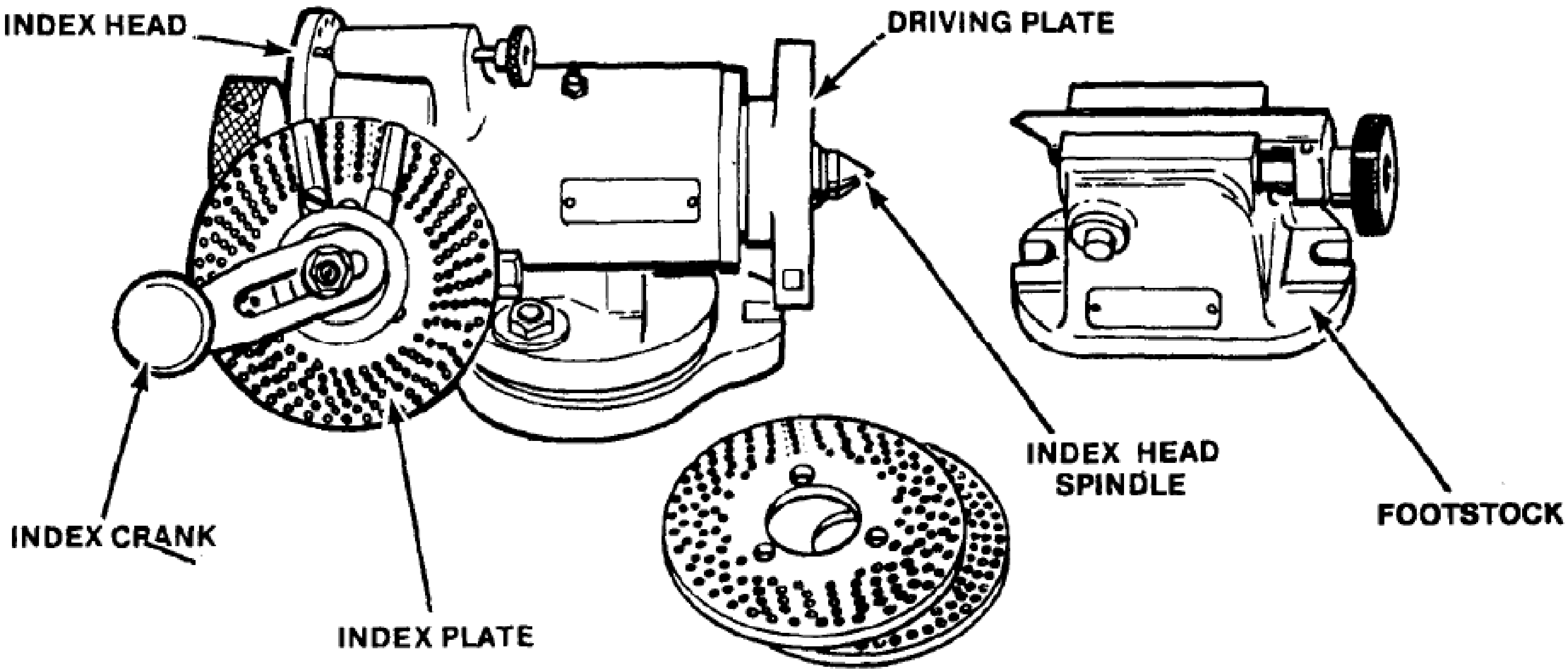
# Ways of workpiece fixation

- **Directly on the milling table**



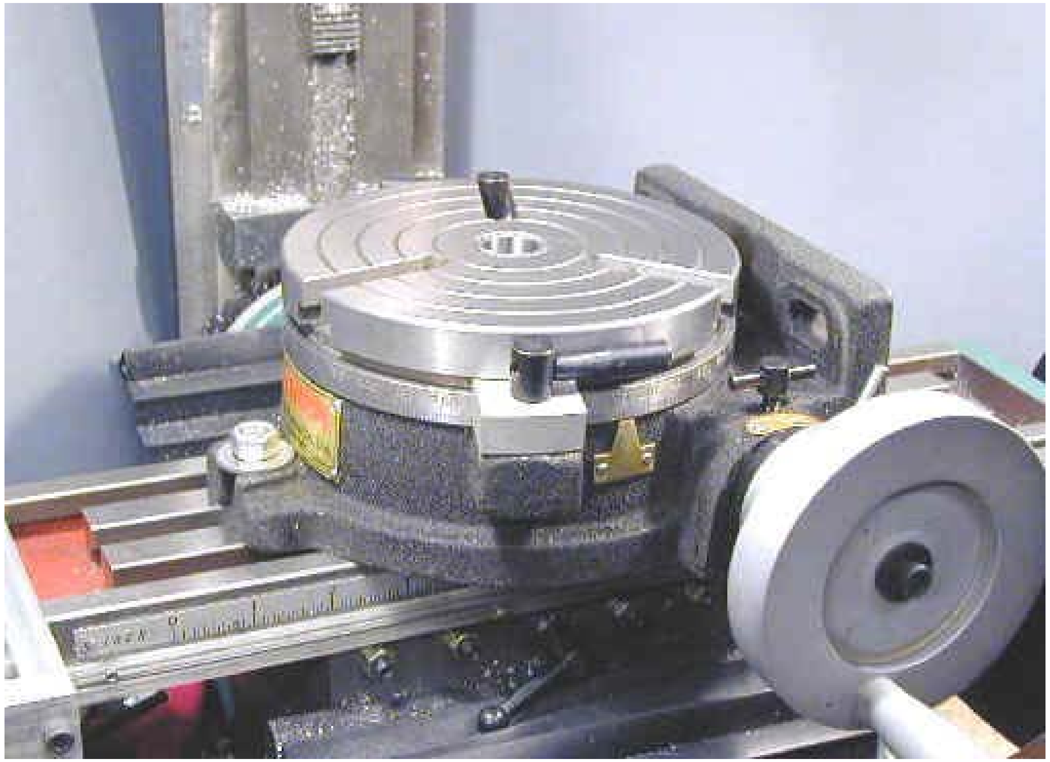
# Ways of workpiece fixation

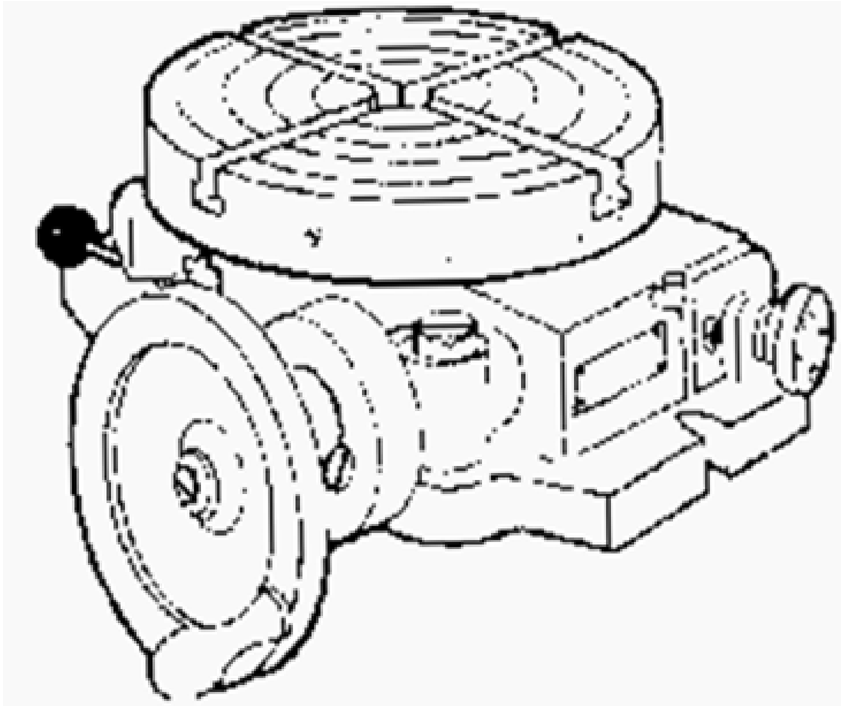
- **Indexing box**



# Ways of workpiece fixation

- **Rotary table**





# Machining Time (T)

$$T = \frac{L}{S * N}$$

- ***Where:***

**T : is the time in minute**

**L : is the travel length of the tool in mm**

**S : is the feed rate in mm / revolution**

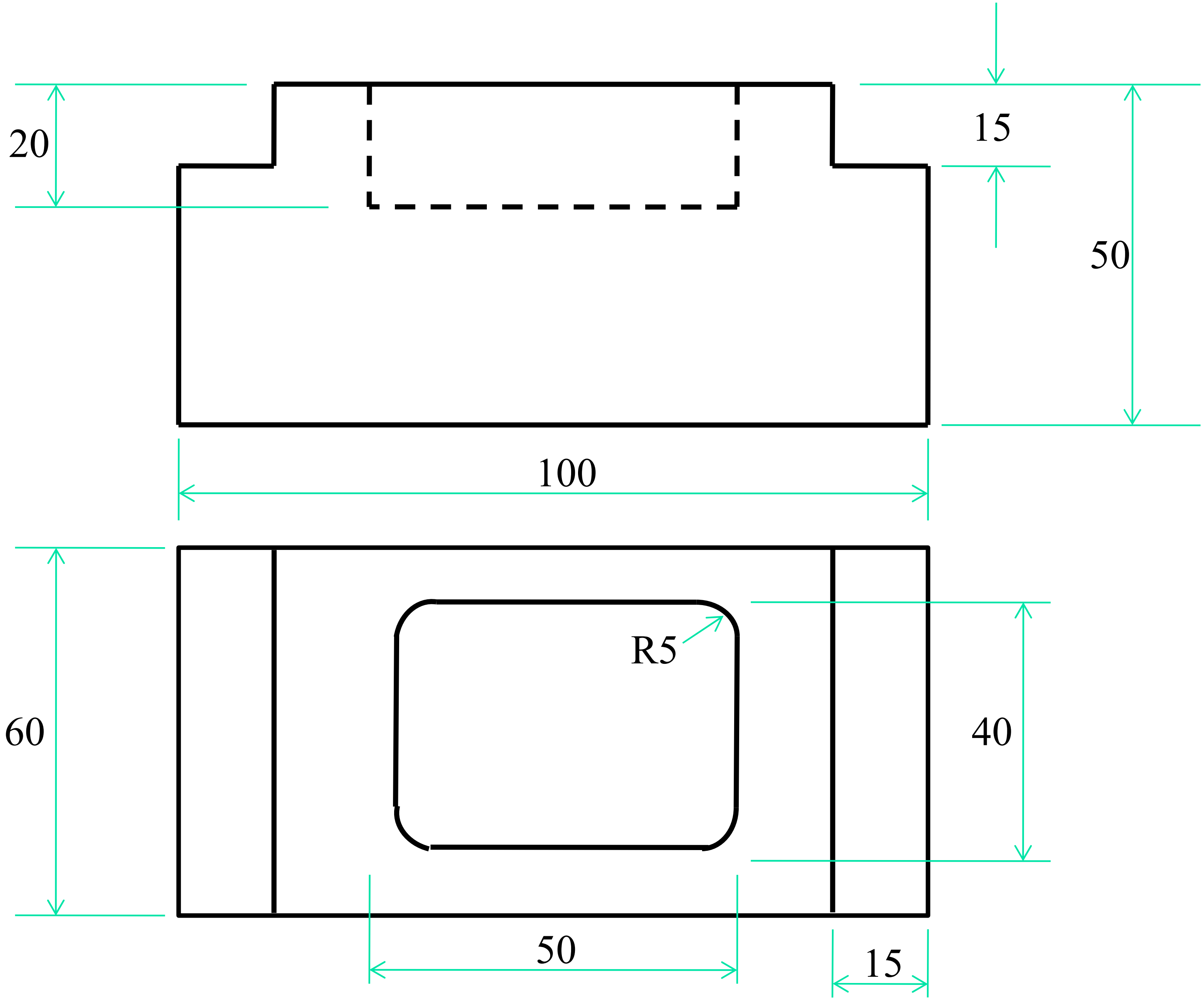
**N : is the rotation speed in rev / minute**

# Machining Time (T)

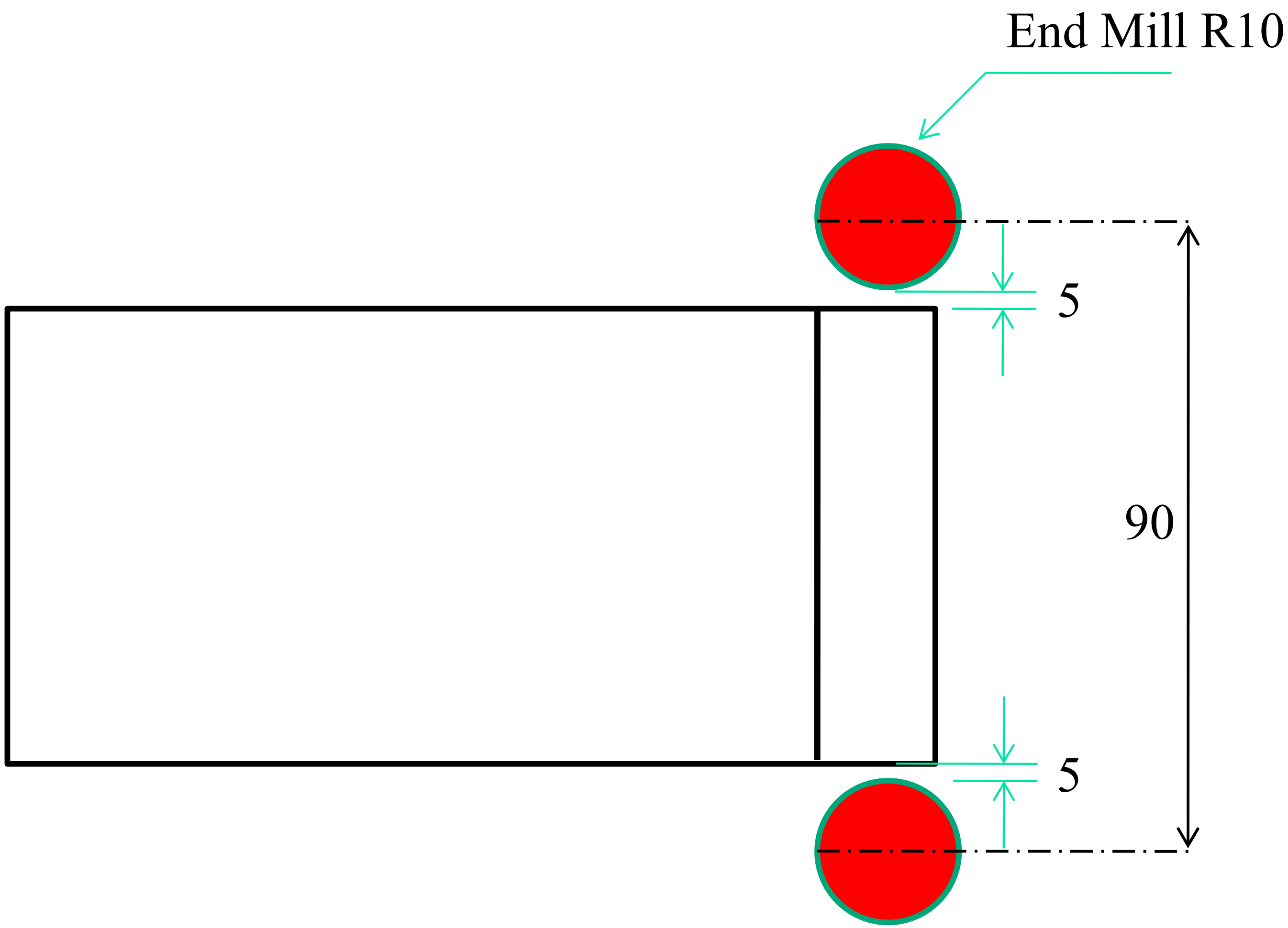
**To solve the problem, you have to determine the following parameters**

- **Sequence of operations**
- **Travel length**
- **Time calculation**

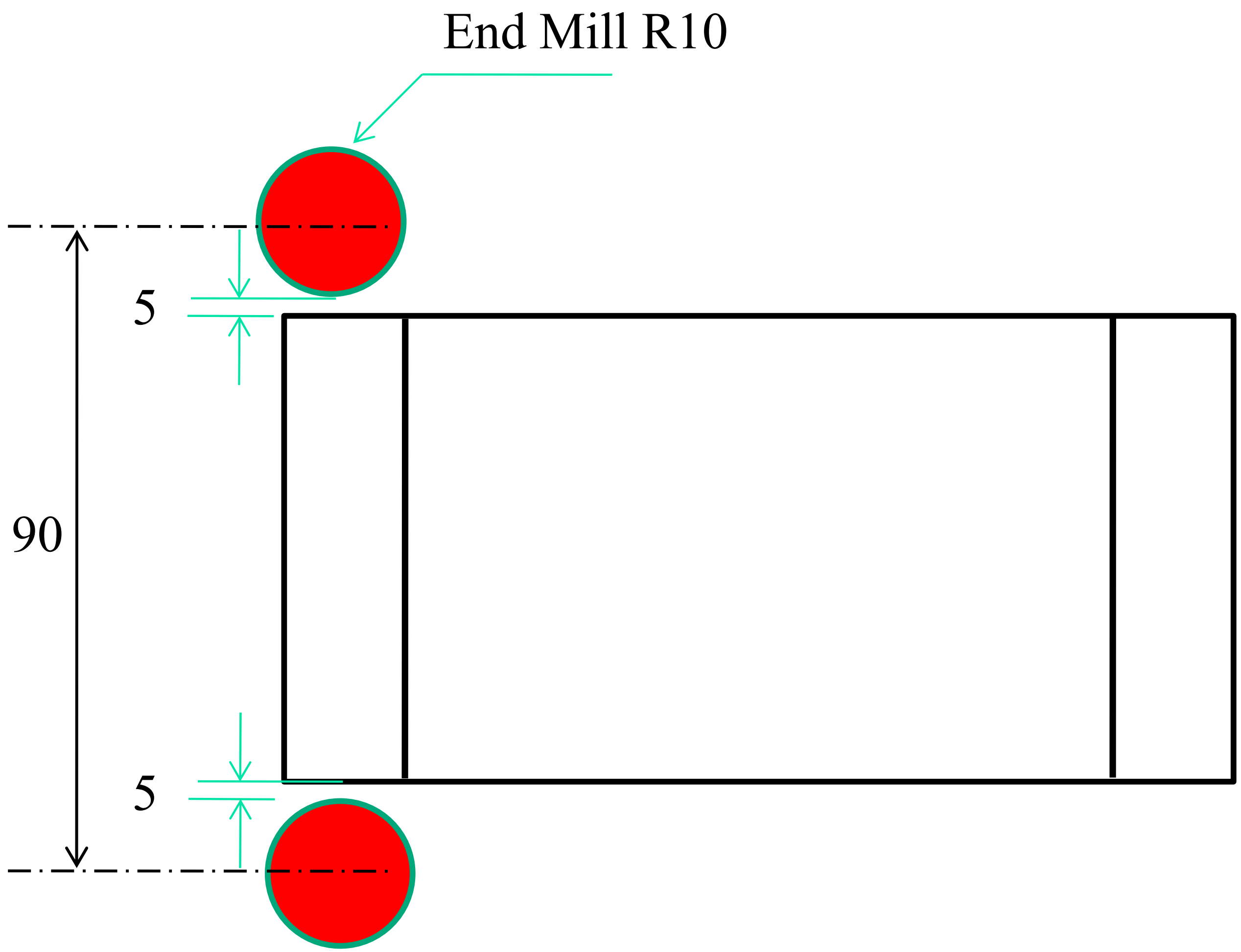
# Exercise



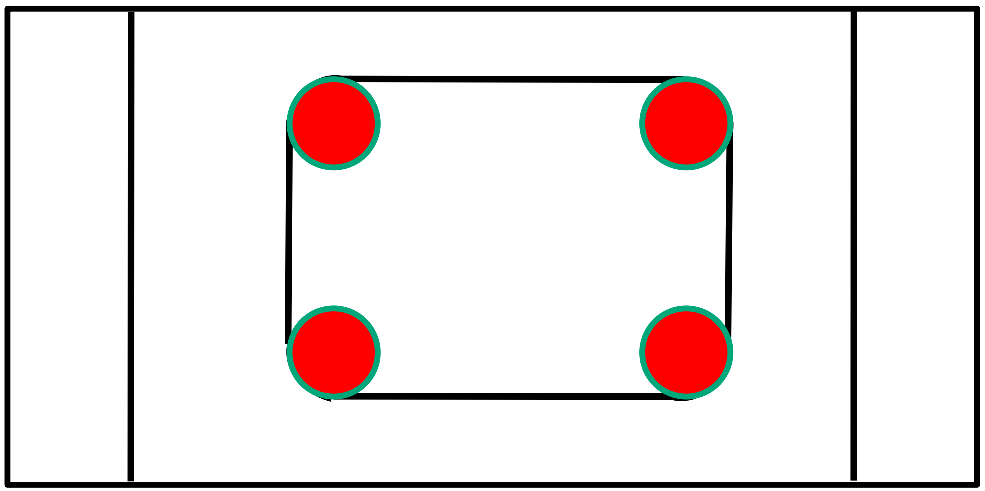
# Sequence of operations



# Sequence of operations



# Sequence of operations



# QUESTIONS?

# THANK YOU